Safety and Its Importance

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1 Safety and Its Importance

Introduction

- In general, safety refers to freedom from danger, risk, threat, harm, damage, injury, or accident. It refers to the minimisation of human-hazard contacts in an industrial setting and is mainly important to reduce the risk of physical harm (injury) to life or property. The condition of firm operations in which accident-free operations are achieved through controlling risks and hazards is referred to as industrial safety.
- Safety is described as a positive, structured activity or programme based on an understanding of the reaction between people and their working environment that supports the business operation by reducing mortality, injuries, illnesses, fires, explosions, and other occupational mishaps. The opposite (antonym) of danger is safety. The absence of probable danger represents total safety. Yet this is a goal that is rarely attained. Protection against risks is what safety is all about.
- A safe situation is one, in which the risks are acceptable. Thus, safety entails lowering or maintaining the hazard level below a legal safe level. It encompasses protection for safety, health, and environment, as well as, property protection.

Safety: Benefits and Needs

- There are numerous advantages to maintaining a safe work environment. Companies that employ people must try to maintain safe workplaces. Injuries on the job are something that prudent employers avoid. They take a moment and use the available resources to ensure the safety of people who come to work every day. Employers need to ensure that their employees or workers reach home in the same condition as they had arrived for work.
- When a workplace is safe, employees feel at ease, leading to increase in productivity and profit. When organisations take steps to develop an effective safety programme, absenteeism decreases. Investing in a robust safety programme has additional advantages. When employees feel safe at the workplace, they are able to serve better. Employee safety has an indirect influence on a company's financial performance as well, for example, customers are ready to switch to another company, if they feel that they are not getting the required services or assistance in time.
- Companies have gone bankrupt while protecting themselves against tort claims alleging wrongful damage or death; legal bills alone can cost exorbitantly. Workplace injuries, as well as, the ensuing lawsuit and insurance cost are frequently avoidable and wasteful expenses. They are distracting and discouraging to employees.
- Workplace safety is paramount because it protects employees and employers from death or injury; teaches them to work in a safe environment; makes everyone feel safe and happy; teaches them to be aware of their surroundings; and protects businesses from civil suits, citations, and fines.
- The formal safety programme is a collection of written documents outlining a company's safety policy goals and obligations. The programme is intended to provide structure and consistency to an organisation's accident prevention activities.

Basic Terminologies and Definitions

Safety Rules

For the protection of each individual and all employees, a list of specified Work Safety Practices must be established. These 'working conditions' can help to prevent accidents during manufacturing. But employees and companies often forget about these, unless enforced.

Specific Written Programmes

- Critical job site threats or risks must also be controlled by providing specialised written programmes, and imparting personnel training, according to the federal and state legislation. These programmes include confined space entry, lockout/tagout, fall protection, scaffolding safety, hazardous substances, and others. Strict protocol must always be followed to avoid exposures, fatalities, and injuries.
- Employers must have a comprehensive safety programme that includes relevant site-specific safety information. It should cover areas like appropriate accidents, accident safety and mitigation promotion, compliance with safety regulations, emergency management, personal protective equipment, safety practices, equipment and machinery, chemical and toxic materials safety, and workplace dangers to name a few.

Benefits of a Training Programme

- An excellent training programme can help to reduce the number of injuries and fatalities, property damage, legal liability, diseases, workers' compensation, and lost work time.
- Safety training classes aid in the establishment of a safety culture, in which workers and employees encourage proper safety procedures when on the job. All staff must be educated as regards to the safety guidelines, and made to understand the significance of workplace safety.

Accident

An accident is an unforeseen event that causes bodily harm or property damage.

Incident

An incident is an unforeseen event that does not cause human injury but may cause property damage or warrant documenting.

Near Miss

It refers to an accident, or an unfortunate incident or event that did not occur but would have occurred had it not been identified. Individual action, established procedures, maintenance breakdown, and a few other factors can all contribute to such incidences.

Hazard

It refers to any building, location, equipment, tool, work, task, or action that poses a substantial harm or risk to employees. Serious injuries are defined as those that cause time loss (beyond a grace period) and/or hospitalisation beyond an emergency department visit. In other words, a hazard is the risk of an injury (or injuries). In practice, a hazard is, generally, related to a disorder or behaviour that, if not regulated, can cause injury or sickness. Hazards are divided into the following categories.

Safety Hazards

- These are the most common and can be encountered at almost any location at some point. These include dangerous situations that can pose a risk, disease, or even death.
- Safety hazards include the following.
  - Spills on surfaces or tripping hazards, such as blocked aisles or cables running across the floor
  - Functioning from heights, including ladders, scaffolds, roofs, or any elevated work area
  - Unprotected factory equipment and moving parts of machines; mechanical parts that a worker can accidentally touch
  - Machinery hazards (lockout/tagout, boiler safety, forklifts, and so on)

Biological Hazards

- Biological hazards are associated with the handling of animals, humans, or infectious plant materials. Working at schools, daycare facilities, academic institutions, hospitals, nursing homes, laboratories, emergency response units, and other settings may expose people to biological hazards.
- People may also be exposed to the following things, such as body fluids and blood, mould/fungi, viruses, plants, and bacteria, excreta from animals and birds, and insect bites, etc.

Physical Hazards

Physical hazards are environmental factors that can cause harmful effects. Physical hazards include ionising and non-ionising radiation (EMFs, microwaves, electromagnetic frequencies, and so on), constant exposure to sunlight/ultraviolet (UV) rays, extreme temperature swings—cold and hot and constant loud noise.

Ergonomic Hazards

- Ergonomic dangers arise when an activity, postures, and working atmosphere put a strain on your body. They are difficult to recognise because you do not always feel the pressure or damage that these threats represent. Short-term exposure may result in 'sore muscles' the next day or in days after exposure. However, long-term exposure may result in significant long-term diseases.
- Ergonomic hazards include harm caused by workstations and chairs that are not correctly fitted, frequent lifting, poor posture, and repeated activities, requiring excessive effort.

Chemical Hazards

- A chemical hazard at a workplace may involve contact with solid, liquid, or gaseous substances. Some people are sensitive to chemicals. Even common solutions can cause illnesses, itching, or difficulty in breathing.
- One needs to be careful while handling liquids, including cleaning solutions, paints, chemicals, and solvents, welding fumes and vapours, and also solvent exposure, especially if the substances are not labelled.
  Examples: some gases, such as acetylene, propane, carbon monoxide, helium; flammable liquids and gases: gasoline, solvents, volatile compounds, and pesticides.

Psychosocial Hazards

- Stressors or dangers that cause stress (short-term repercussions) and strain are referred to as psychosocial hazards. These have adverse long-term effects. These are the dangers associated with workplace issues, including excessive workload, loss of control and/or respect, and so on. Workload demands are one type of psychosocial risk.
- Psychosocial hazards include workplace violence, degree of difficulty and/or speed (or lack of), adaptability, controlling or saying things, social support/relationships, and sexual harassment.

Risk

- It is related to a hazard. It is the chance of injury or loss (typically economic) as a result of a potential hazard and its impact on a firm or an individual. Risk is often associated with medical conditions or exposure, and consequently focuses on employee health.
- Calculating risk: the risk posed by a certain hazard can be thought of as growing in three ways.
  - With the possibility that a dangerous event will occur
  - With exposure to that event
  - With the potential consequences of that occurrence
- Each of these three elements is assigned a numerical value for risk calculation. The product of these three criteria is then used to construct an overall risk score. The numerical representations, while set randomly, are self-consistent, and when combined, they produce a realistic yet relative score for overall risk.

$$R = C \times E \times P$$

Where, C is the consequence rating.
E is the exposure value.
P is the probability value.

Risk Management

Risk management is a critical procedure that involves preventing mishaps and accidents from occurring. It is accomplished by evaluating risks based on safety assessments and significance and frequency of previous safety-related events, implementing suitable steps corresponding with risk levels, and maintaining risks within acceptable bounds.

Management Guidelines for OSHA Programme

The Occupational Safety and Health Administration (OSHA) guidelines provide an appropriate and adaptable foundation for addressing workplace safety and health issues for employers, workers, and others present at the workplace. They are intended for use in any environment but will be incredibly beneficial in small- and medium-sized businesses.

Management Leadership

Management is defined as the structure, direction, and resources required for the development of an effective safety and health system. Business owners, CEOs, supervisors, and managers at all levels comprise the management leadership. The leadership is devoted to continuously improve the occupational health and safety standards—making the health and safety of workers paramount to an organisation, making available appropriate resources to ensure a safer workplace, and setting high standards by using the appropriate safety equipment, when necessary.

Employee Participation

Workers stand to gain the most from a good 'safety and health programme', and they stand to lose the most if it fails. Workers are often aware of their jobs and the potential hazards. Workers and (if relevant) their leaders must actively participate in the formulation and implementation of a safety and health programme.

- All workers, including contractors, freelancers, and temporary personnel, are eligible for participation.
- They must engage in programme design and implementation, and have access to data required to effectively participate in the programme.
- They must be encouraged to join the programme and report any health/safety concerns.

Hazard Identification and Assessment

- A proactive, ongoing approach for identifying and analysing danger is necessary for an effective safety and health programme. Failure to recognise or identify hazards is frequently one of the 'underlying causes' of work related casualities, indicating an ineffective health and safety programme. Hazard identification can lead to opportunities that increase programme performance.
- To detect and assess hazards, one must:
  - gather information on potential hazards at the workplace, and inspect the workplace regularly to discover new or reoccurring dangers.
  - look at injuries, illnesses, mishaps, and close calls/near misses to determine what caused them.
  - recognise and handle trends in reports of injuries, illnesses, and hazards, and think about the risks associated with emergency or quasi-scenarios.
  - assess the impact and frequency of incidents that might occur as a result of each identified danger and use this knowledge to prioritise corrective activities. Employers, on the other hand, have an ongoing obligation to control major identified hazards.

Control and Prevention of Hazards

- Preventive measures are important to protect people from general workplace hazards, such as accidents, illnesses, and other health issues. Employers must try to provide a healthy and safe working environment.
- The measures indicated in this section will help you prevent and control the hazards listed in the previous step of these guidelines.
- The following steps are involved in hazard prevention and control.
  - Assess the risk control measures.
  - Use a hazard management framework to guide remedy selection and implementation.
  - Develop actions to protect staff in the event of a crisis.
  - Evaluate the effectiveness of controls to determine whether more or alternate controls would be more effective. Emerging innovations may be more reliable, dependable, or cost-effective to operate.

Instruction and Training

- Workers, aware of workplace hazards and the controls in place, can operate more safely and productively.
- Employers, managers, supervisors, and workers must have the required information and skills to work properly and avoid creating situations that could endanger life and property.
- Education and training show that you are aware of and understand workplace risks and hazards, as well as, know how to recognise, report, and handle them. One must get specific training, if one's job includes handling unusual hazards. Furthermore, when specific duties in managing or executing the safety and health programme are allocated, all staff receive specialised training. This training is required to ensure that employees can fully participate, as indicated in the guidelines 'Worker Participation' section.

Evaluation and Enhancement of Programmes

- Take a step back from time-to-time to assess what works and what does not, if you are on schedule to meet your programme objectives. Once the programme is in place, evaluate it regularly to ensure that it is being implemented as intended. When you find possibilities to improve the software, make changes and track how well it operates. Share your evaluation and monitoring results with your co-workers to assist the drive further.
- For the evaluation and enhancement of programmes you should:
  - create, monitor, and analyse metrics that show whether the programme is effective as part of programme evaluation and improvement.
  - conduct initial and periodic evaluation of the programme to identify shortcomings and possibilities for improvement.

Occupational Health and Safety

- Occupational health and safety is a science that encompasses many specialised professions. It includes workers' social, emotional, and physical well-being. It comprises the ergonomic, physical, chemical, biological, psychological, and social components of occupational safety and health. The discipline of occupational health and safety is concerned with the preservation and protection of human resources at the workplace.
- It consists of the following components.
  - Promotion and maintenance of the highest level of mental, physical, and social well-being of workers in all occupations
  - Prevention of health deviations among workers caused by the working conditions, and protection of workers from risks that may adversely affect their health
  - Placement and maintenance of a worker in an occupational environment adapted to his/her physiological and psychological equipment

Occupational Health and Safety Goals

- The primary goal of occupational health and safety on a worldwide scale is to ensure that every worker in any sector of the economy has access to health and safety. Since its foundation, the World Health Organization (WHO) has included components of occupational health in all its policies. The need of protecting workers from occupational health hazards and promoting workplace safety for all has been stressed in key WHO documents.
- According to the paper, 'Global Strategy on Occupational Health for All,' the high-priority objectives are as follows.
  - Strengthening international and national regulations for workplace health and providing appropriate policy tools
  - Creating an appropriate work environment
  - Creating healthy work practices and promoting workplace health
  - Creating occupational health support services
  - Creating occupational health standards backed by scientific risk assessment
  - Fostering human resource development for occupational health
  - Creating registration and data systems, creating information services for experts, efficient data transmission, and generating public awareness through public information

- Conducting research studies in the area
- Increasing cooperation in occupational safety and other activities and services

Workplace Hazards and Hygiene

Occupational Hazards

- An occupational hazard is any condition that may have a negative impact on the well-being or health of those who are exposed to it. Hazardous agents and workforce potentially exposed to these must be identified at a workplace.
- Occupational hazards are classified into two types: safety hazards that cause accidents that physically damage workers, and health hazards that cause allergies or diseases. It is critical to understand that a 'hazard' only represents the potential harm. The extent and length of exposure, as well as, the number of personnel exposed, will determine if it causes harm. The occurrence, or severity of occupational sickness is concerned with circumstances encountered on the job or at the workplace.

Occupational Hygiene

The discipline of forecasting, recognising, analysing, and controlling health hazards, and risks to protect workers' health and well-being, and secure the community at large is defined as occupational hygiene. As a result, it includes recognising dangerous agents at the workplace (chemical, physical, and biological) that may cause illness or discomfort. It also includes determining the level of risk as a result of exposure to these harmful substances, and risk management to prevent illnesses, whether in the short or long term.

Occupational Hygiene Practice

- The steps that need to be taken care of for ensuring industrial hygiene practices are as follows.
  - Become aware of potential health concerns at the workplace.
  - Hazard evaluation is the process of analysing exposure and determining the amount of risk to life and property.
  - Hazard prevention and control is the process of designing and executing plans to eliminate or limit the incidence of hazardous agents and factors at the workplace while also considering environmental protection.
  - Contribute to the entire risk assessment and management of an agent, process, or workplace, as well as, the formulation of risk management priorities.
  - Understand the legal foundation for practising occupational hygiene.
  - Educate, train, inform, and advise people at all levels about hazardous situations and eventualities involved before setting up a new workplace.
- An optimal approach to hazard prevention should include the following.
  - Occupational health and environmental impact studies
  - Choosing the least hazardous, least damaging technology (cleaner production)
  - An environment friendly site
  - Appropriate design, including layout and control technology, as well as, safe treatment and disposal of the resulting industrial effluent and waste
  - Development of legislations and guidelines for training on process operation, including safe work practices, maintenance, and emergency plans

Ergonomics

- The application of scientific knowledge about humans to the design of items, processes, and environments for use is known as ergonomics. The goal is to ensure that the job fits the person in every way and that the work setting does not impair human talents and limitations. As a result, ergonomics is applied to anything that concerns people. If correctly planned, workplaces, sports, leisure, health, and safety should all incorporate ergonomic concepts.
- Ergonomics, in general, tries to enhance the fit between work, workers' physical abilities, information handling, and workload capacity. In North America, the subject is known as human factors engineering.
- Workstation ergonomics aims to fit the workstation to the employee, minimise awkward positions, promote proper posture,

and recognise potential threats, such as excessive glare, loudness, and so on.

Fire Safety Engineering

Fire Physics and Chemistry

- Fire is a complex chemical process, and emergency responders must grasp the underlying chemistry and physics to establish judgements based on scientific principles. The diffusion flame process (fire) is made of three fundamental components: fuel, oxygen, and heat.
- Dawson Powell defines the six aspects of fire life cycle as input heat, fuel, oxygen, composition, blending, and ignition continuity. All of these ingredients are required for the diffusion flame combustion process to begin and continue. The fire triangle represents the initial three elements: input heat, fuel, and oxygen.

Element 1: Heat Input

- Materials, whether solid or liquid, do not burn. These materials must be heated enough to form vapours before combustion can occur. Their fumes burn. The flashpoint is the lowest temperature at which a solid or liquid material emits enough vapour to burn under laboratory circumstances. The flame point is a few degrees above the flashpoint, i.e., the temperature at which the fuel will continue to generate enough vapours to sustain a continuous flame.
- The ignition temperature is the temperature at which the vapour will ignite. Piloted ignition occurs when the source of heat is an open flame or spark.

Element 2: Fuel

- Fuel at room temperature can be in the form of a gas, liquid, or solid. As previously stated, liquid and solid fuels must be sufficiently heated to produce vapours. In general, combustible implies capable of burning, often in the air under normal temperature and pressure circumstances, whereas flammable means capable of burning with a flame. This is not to be confused with flammable and combustible liquids.
- Flammable liquids, such as gasoline, acetone, and ethyl alcohol, have flashpoints below 100°F (37.8°C). Combustible liquids, such as kerosene and fuel oil, have a flashpoint at or above 100°F (37.8°C).

Element 3: Oxygen

- The air in the atmosphere provides oxygen to the fire triangle. Air is composed of approximately 79 per cent nitrogen and 21 per cent oxygen.
- The OSHA defines a hazardous environment as one that is oxygen deficient (less than 19.5 per cent oxygen) or oxygen enriched (more than 23.5 per cent oxygen). The OSHA considers either situation to be an environment immediately hazardous to life and health (IDLH) for issues unrelated to the presence of fire. Depending on the type of fuel used, fires can occur with far less oxygen present than required to support respiration.

Elements 4 and 5: Mixing and Proportioning

- Mixing and proportioning are actions that must occur continuously for fire to spread. The precise proportions of fuel vapour and oxygen must be combined. A combination of fuel vapour and oxygen of this type is said to be explosive or flammable. The explosive or flammable limits are stated as a percentage of the concentration of fuel vapour in the air. A combination that includes less gasoline vapour than required for ignition is too lean, whereas a mixture that includes too much fuel vapour is too rich. The Lower Explosive Limit (LEL) is the lowest concentration that will burn, while the Upper Explosive Limit (UEL) is the highest concentration.
- The weight of a volume of a given gas to an equal amount of dry air, where the air is assigned a value of 1, is another essential attribute of gases. A vapour density less than 1 indicates that the gas is lighter than air and will rise in a generally quiet atmosphere, whereas a vapour density greater than 1 indicates that the gas is heavier than air and will sink to the ground.

Element 6: Continuity of Ignition

- The heat feedback from fire to the fuel is referred to as ignition continuity. Conduction, convection, radiation, and direct flame contact—all transport heat.
- Conduction is the direct passage of heat through a solid mass. A hot burner, for example, is conducted through the pot to its contents.
- Convection is the transfer of heat by the motion of hot materials, such as smoke, hot air, heated gases created by fire, and embers in flight.
- Radiation is the transport of heat from a source across an intervening space without the use of any solid substance.

Fire Properties of Solid, Liquid, and Gas

- Most materials must be in gaseous or vapour state to be ignited. These gases or vapours must be present in sufficient amount in the atmosphere to generate a combustible mixture. The generated fuel vapour should then be heated to ignition temperature.
- The time and energy necessary for ignition are determined by the ignition source's energy, the thermal inertia (k, p, c) of the fuel, the minimum ignition energy required for that fuel, and the geometry of the fuel. If the fuel is to achieve its ignition temperature, the rate of heat transferred to it must be greater than the rate of conduction into or through the fuel, as well as, the losses due to radiation and convection. Cigarettes, upholstered furniture, sawdust, and cellulosic insulation are permeable and allow for easy air infiltration. These materials can smoulder and burn as solid phase combustion. Smouldering is dangerous because it produces more poisonous chemicals per unit of mass consumed than blazing combustion, and it allows for fiery ignition from a heat source that is too weak to ignite a flame directly.
- Smouldering is the incorrect name for a non-flaming response of solid fuel to an exterior heat flow. When exposed to adequate heat flux, solid fuels, such as wood will decompose and emit vapour. This gasification process is frequently endothermic as there is little or no oxidation involved. This is called forced pyrolysis, and not smouldering.
- The quantity of heat transmitted per unit area per unit time from or to a surface is defined as heat flux.

Ignition of Solid Fuels

- To burn with a flame, solid fuels must be either melted and vapourised (as with thermoplastics), or pyrolysed into gases, or vapour (i.e., wood or thermoset plastic). To generate vapour, heat must be applied to the fuel in both cases. The time or energy required for solids to ignite varies on many factors, which are as follows.
    - Material density: high-density materials of the same basic type (wood and plastic) transmit energy away from the ignition source faster than low-density materials, which act as insulators and allow energy to linger at the surface, for example, low-density foam plastic, ignites faster than high-density plastic.
    - Surface area to the mass ratio: the surface area for a given mass influences the amount of energy required for ignition. It is quite easy to fire one pound of thin pine shavings with a matchstick, but it is unlikely to burn a one-pound solid piece of wood with the same matchstick.
    - For spontaneous ignition to arise as a result of radiative heat transfer, the volatile compounds emitted from the surface must be hot enough to generate a combustible combination when it mixes with unheated air above the auto-ignition temperature. In contrast with convective heating, the air is already at a high temperature. Thus, the volatiles do not need to be as hot. Thin solids ignite faster than thick solids when exposed to their ignition temperature (e.g., paper versus plywood).

Ignition of Liquids

- A liquid's vapour must be at or above its flashpoint to generate an ignitable combination.

Based on an applicable ASTM test method, the flashpoint of a liquid is the lowest temperature at which it emits enough vapour to support a brief flame across its surface. The flashpoint value may differ based on the type of test done. Even if the majority of a liquid is somewhat below its flashpoint, an ignition source can generate a locally heated patch large enough to cause ignition.

- Atomised liquid or mist (those with a high surface area to mass ratio) can be ignited more easily than in bulk. In case of sprays, ignition can frequently occur at ambient temperatures below the bulk liquid's recorded flashpoint, if the liquid is heated above its flashpoint and ignition temperature is somewhere near the heat source.

Ignition of Gases

Gaseous combustible compounds have incredibly low mass and demand the least amount of energy to ignite.

Industrial Fire Protection Systems

Sprinkler

A fire sprinkler system consists of a water supply network that provides suitable pressure and flow to the water supply pipes that connect the fire sprinklers. Fire sprinkler systems can significantly minimise the heat, flames, and smoke caused by a fire because they react very fast. Although primarily employed in industries and major commercial structures, sprinkler systems for houses and buildings are now affordable.

Fire Hydrants

- A fire hydrant is a conduit that permits the flow of water from the water main via a valve to extinguish a fire. It is the foundation of a firefighting system in a building or premise.
- A water distribution system consists of a water tank, suction pipes, fire pumps, and a dispersed piping system. The dispersed plumbing system connects fire hydrants, hoses, and valves throughout the building.
- The fire hydrant system's objective is to offer an easily accessible stream of water at any point throughout the building. This aids in fire control during an emergency. The firefighter is given a dependable and adaptable system capable of using various techniques to control the fire via the fire main system. Water can be delivered as a continuous stream for fighting deep-seated flames and as a spray for fighting flammable liquid fires via the fire hydrant system.

Standpipe

- Standpipe systems are merely an extension of the fire hydrant system, consisting of many pipes that link water supply to hose connectors. They are intended to supply building inhabitants or the fire department with a pre-piped water system. Standpipe systems are intended to supply water to hose lines in crucial areas in a building.
- Some older buildings have only standpipe systems; however, many contemporary structures have a hybrid system that supplies both the fire sprinkler and the standpipe systems. They are most commonly found in large floor area buildings, where the majority of the facility may be located at some distance from the entrance and in highrises to avoid large hose length in stairways and on ground.

Deluge

- The deluge system is designed to protect high-risk regions featuring a significant fuel hazard with a high heat release rate by simultaneously activating a large number of open sprayers in the event of a fire. They are sometimes used to slow the spread of fire in personnel egress channels or building openings (e.g., openings in a fire-rated wall).
- Deluge systems are linked to a water supply via a deluge valve that is activated by smoke or heat-detecting system. The sensor system is mounted alongside the sprinklers. Water is discharged through each of the sprinkler heads in the system when the detecting system is engaged.

- Deluge systems are employed in high-risk environments, such as power plants, hangars, and storage or processing facilities. Deluge systems are required where rapid fire suppression is required to check the spread of fire.

Emulsifier

- An emulsion is a combination of two or more typically immiscible liquids (unmixable or unblendable). Emulsifying agents are excellent at putting out fires caused by modest, thin-layered spills of combustible materials. These agents encase the fuel in a fuel-water emulsion, confining the combustible vapour in the water phase. An aqueous buffer solution is applied to the fuel via a high-pressure nozzle to create this emulsion.
- Emulsifiers are ineffective at extinguishing massive fires involving bulk/deep liquid fuels because the amount of emulsifier agent required is proportional to the volume of the fuel, while other agents, such as aqueous film-forming foam (AFFF), require only to cover the surface of the fuel to achieve vapour mitigation.

Carbon dioxide (CO_2)

- CO_2 is a colourless, odourless, electrically non-conductive gas that is extremely effective as a fire suppressant. The mechanisms that cause CO_2 to extinguish flames are widely understood. Returning to the traditional fire triangle, we see that the combination of fuel, oxygen, and heat is required to establish a fire situation.
- Carbon dioxide extinguishes fire by targeting all three points of the fire triangle. The oxygen level of the atmosphere is the major target. The injection of CO_2 into the fire zone displaces enough oxygen in the atmosphere to put out fire. Simultaneously, a drop in the percentage of gasified fuel in the fire region aids the extinguishing process. Finally, CO_2 provides some cooling in the fire zone, which helps to complete the extinguishing process.

Firefighting Foam

- Firefighting foam is made of three components: foam concentrate, water, and air. When mixed, these components generate a uniform foam blanket that extinguishes fire through a combination of cooling, removing the flame origin from the product surface, suppressing fumes, and smothering mechanisms.
- Foam suppression systems are an excellent way to prevent flammable and combustible substances.

Dry Chemical Powder (DCP)

- Dry chemical is a powder made of minute particles of sodium bicarbonate, potassium bicarbonate, urea-based potassium bicarbonate, or mono ammonium phosphate with additional particulate material.
- DCP provides packing resistance, moisture absorption resistance (caking), and adequate flow capabilities.

Halogenated Agents

- Halons are chemical compounds of methane (CH_4) or ethane (C_2H_6), in which some or all of the H atoms are substituted with halogen atoms, such as fluorine (F), chlorine (Cl), or bromine (Br), or mixtures of these halogen elements.
- Halogenated extinguishing agents, although a relatively new discovery in fire defence, are already being phased out due to their extremely high ozone depletion potential (ODP) (power for depleting the ozone layer above the Earth's surface, which acts as a shield for protecting living beings from harmful ultraviolet rays).
- Halon 1301 (Bromo-trifluoro-methane, CF_3Br) is the most often used halon because it has the lowest toxicity and maximum extinguishing efficacy. Most of the fires could be doused by 4–6 per cent volume of H1301.

Fire Load of a Building

- A building's or compartment's fire loading is a method of determining the probable intensity

of a hypothetical fire. It is the heat production per unit floor area, generally represented in kJ/m^2, estimated from the calorific value of the materials present, or it can be stated in terms of the weight of flammable building contents per unit building floor area.

- Fire loads include all flammable building items, such as furnishings and equipment, as well as, combustible building components.
- Typically, the majority of the fire load in a building is caused by contents introduced after the building construction has been completed.
- Fire loading is a method of assessing industrial safety concerns.

Classification of Fires

According to the type of the fire materials, the Bureau of Indian Standards has categorised fires into the following categories:

- Class A fires: these are fires involving solid flammable materials, such as wood, paper, cloth, plastic, rubber, and so on.
- Class B fires: these fires contain flammable liquids, such as oil, grease, gasoline, paint, varnish, thinner, and so on.
- Class C fires: these are the fires that include gaseous substances under pressure, such as liquified petroleum gas.
- Class D fires: these fires involve metals, such as magnesium, aluminium, lithium, zinc, sodium, potassium, and so on.
- Class E fires: these are fires in electrical devices, such as generators and transformers.

Safety in Construction

- The construction industry is second to agriculture in terms of its contribution to the Indian economy. It is a labour-intensive industry that employs a sizable rural population. At the same time, it is one of the most accident-prone industries, with a relatively high frequency of work-related accidents. In India, where regulations and procedures are frequently ignored, fatal building accidents are fairly prevalent. Because the business is so fragmented, there exist no statistical data on occupational injuries and deaths in India. There is no reliable system in place, nor is there any organisation to monitor accident rates. Workers in this business are usually uneducated, who are unaware of the risks to which they are subjected and are also unaware of their rights. This makes it easier for contractors to exploit them and increase their profits.
- As a result of globalisation, an increasing number of international corporations are undertaking infrastructure projects in India. Many of these multinational corporations have made it a policy to prioritise health, safety, and environment. Construction is a high-risk industry that includes a wide range of tasks, such as building, altering, and/or repairing. The industry involves the construction of residential buildings, bridges, roadways and pavements, and also excavations, demolitions, and substantial painting operations. Construction workers are exposed to numerous hazards, including falls, accidents caused by machinery, being hit by heavy construction equipment, electrical shocks, and allergies caused by silica dust, and asbestos.
- Falls, being stuck between things, electrocution, and being hit by an object are some of the most common safety hazards at construction sites. Such incidents have led to causing injuries and deaths at construction sites around the world. Failures in danger detection are frequently caused by inadequate or ineffective supervision and training of construction workers. Tasks in design for safety, safety inspection, and monitoring safety are examples of fields with low training. Failure in either of these areas can pose a threat to construction workers.
- Most accidents are caused by a combination of the following.
    - The management's inability to put in place and implement measures to safeguard workers from potential hazards at the workplace
    - Numerous risky acts undertaken by workers themselves
    - Construction project characteristics, such as project nature and approach

Major Hazards and Risks

The following are the primary hazards and in the construction industry.

- Slips, trips, and falls
- Trenching and excavation risk
- Construction machinery risks
- Electric shocks and arc flash/arc explosion
- Inadequate and inappropriate use of personal protection equipment (PPE)
- Repetitive motion injuries

Slips, Trips, and Falls

Slipping is the main cause of accidents in all areas, from heavy industry to office labour. Tripping and falling have also been recognised as major causes of workplace accidents, resulting in more than three days absence. Slips, stumble, and falls can cause severe injuries, such as fractured bones or concussion. Construction slips, trips, and accidents are frequently caused by uneven surfaces, obstacles, trailing wires, moist or slick surfaces, and changes in level. Contractors and others in charge of construction sites must organise work so that people can move around the site safely.

Work at Height

- Every year, falls remain the most common cause of fatalities in the construction industry. Falls are, typically, caused by a mix of factors, including unstable working surfaces, improper or non-use of fall protection equipment, and human error. According to studies, guard rails, fall prevention systems, safety nets, covers, and restraint devices can prevent major injuries and deaths caused by falls. Residential construction workers, who are 6 feet or more, must be equipped with conventional fall protection (such as guard rail systems, safety net mechanisms, or personal fall arrest systems) at all times while working at the site.
- These hazards might arise at any building site. The consequences are frequently more severe as the falling altitude increases. Falls from unsecured edges or gaps at height, roof ladders, temporary platforms like tower scaffolds, and stairways, and during excavations, or working with fragile materials are the most prevalent causes of falls.
- It is vital to assess the risk of the worksite and to organise and plan the task for it to be completed safely. Work at height management adheres to a control system that avoids, prevents, and arrests.
- The minimum health and safety requirements for working at height are as follows.
    - Consider using aerial lifts or elevated platforms to provide safe and secure elevated working surfaces.
    - Install control line safeguards or erect guard rail systems with toeboards and warning lines to assist workers near the periphery of floors and roofs.
    - Cover floor holes; and/or use safety net systems or personal fall arrest systems (body harnesses).
- Scaffolding or other comparable equipment erection and dismantling scaffolding risks can be split into two categories.
    - Those involved in construction, modification, and dismantling of scaffolding
    - Those involved in the usage of scaffolding (e.g., risk of slipping)
- These risks are possible wherever scaffolds are used. Similar hazards may arise when scaffolding-like structures are employed in falsework (temporary constructs used in construction works for holding spanning or arching buildings until the component's development is advanced enough to support itself).
- The incidents that occur are primarily the result of the following.
    - Unsafe procedures of constructing or disassembling
    - Scaffolding flaws where the tower structure is improperly erected, or a platform guard rail is missing
    - Scaffold mishandling when a ladder is used on a tower, causing it to topple, or when someone falls when the tower is being relocated
- The following are the minimal safety and health criteria for scaffold workers while using work equipment.

- The scaffold must be stable, stiff, and strong enough to support its weight, plus four times the maximum designed load without sinking or shifting. It must be built on a solid ground.
- Stable objects, such as barrels, crates, loose bricks, or concrete blocks, should not be utilised to support scaffolding or planks.
- Scaffolding must be erected, relocated, dismantled, or altered only under the supervision of a competent person.
- Guard rails, mid-rails, and toeboards must be installed on scaffolds.
- Scaffold components, such as bracing, brackets, trusses, screw legs, or ladders that are broken or weakened must be repaired or replaced immediately.
- Scaffold platforms must be securely planked with quality scaffold plank or any other suitable material.
- The scaffolding must be inspected and reinspected at regular intervals by a 'qualified person.'
- Rigging on suspension scaffolds must be examined by a competent person before every shift and after any occurrence that may jeopardise structural integrity to confirm that all links are tight and that no damage has occurred since its last use.
- Heat emitting sources must be avoided when using synthetic or natural rope in suspended scaffolding.
- Workers must be educated on the risks of using diagonal bracing for fall protection.
- Scaffolds can be accessed by ladders and stairwells.

- Scaffolds must always be at least one foot away from power lines.
- Erection and dismantling should use the following.
  - Advance guard rail system, in which temporary guard rail modules are fixed in place from the platform level and moved up to it (they are installed before the operator gains access to the platform to accommodate the permanent guard rails)
  - Through the trap, the operator assumes a working position in the platform's trap door, from which one can add or remove the elements that serve as guard rails on the level above the platform. It is intended to prevent the operator from standing on an unprotected platform.
- To maintain scaffold stability, there is a need to make sure that the tower rests on a strong, flat ground with locked castors or base plates appropriately supported. Bricks or building blocks are not used to take the load off any part of the tower; stabilisers or outriggers are fitted when required by the instruction manual. A tower is never erected above a recommended height.
- Using and relocating scaffolds should never be done during high winds, while persons or materials are on the tower, as a platform for ladders, trestles, or other access equipment, with broken or missing pieces; or with incompatible components.
- When using a scaffold, one should always ensure the following.
  - Damaged pieces that impair the stability of the scaffold must be removed during relocation.
  - Scaffolds are not changed. All scaffolds must be completely planked.
  - Scaffolds are not shifted horizontally while workers are on them unless they are meant to be moved. Besides, the workers concerned must be given appropriate and adequate education on the same.
  - Workers are not allowed to work on scaffolds that are blanketed with snow, ice, or other slick materials.
  - No scaffolding is placed or relocated within one feet of power lines.
  - Workers are not permitted to work on scaffolds during inclement weather conditions, or during severe winds.
  - No ladders, boxes, barrels, buckets, or other improvised platforms are used to increase work height.
  - Extra material is not permitted to accumulate on scaffold platforms. Scaffolds should not be stacked with more weight than their design capacity allows.

- The height of a scaffold must be limited to a maximum of 4 m. Ensure that the terrain is firm, level, and devoid of potholes. It must be noted that the scaffold is always pushed or pulled only from the base.

- Falls on the same level are typically caused by slipping and tripping. They are more likely in unkempt areas.



Fig. 1.1

Fall Protection Training

Employers must impart adequate training to workers, who may be exposed to fall hazards. Fall hazards must be identified and minimised during training. When the system is used, each worker is responsible for ensuring the upkeep of the safety monitoring system. However, the employer must ensure that each employee receives requisite training from a qualified supervisor and is made aware of the following points.

- The nature of workplace fall hazards
- The processes for building, maintaining, dismantling, and evaluating fall protection devices that need to be employed
- The usage and functioning of restricted access zones, guard rail, personal fall arrest, safety net, caution line, safety monitoring systems, and other safety measures that need to be implemented
- The restrictions on the use of mechanical devices when performing roofing work on low-slope roofs
- The recommended practice for handling and storing equipment and supplies, as well as, the installation of overhead protection
- Workers' role in fall protection strategies

Risks Related to Construction Machinery

- Construction machinery risks vary depending on the type of equipment (earthmoving equipment, lifting equipment, etc.) used and job roles. Risks associated with earthmoving equipment (backhoes, loader shovel excavators, etc., and their attachments) include equipment rollover, objects dropping onto the equipment, malfunctioning safety, other warning mechanisms, and so on. Workers falling from a height during the installation, operation, and dismantling of lifting equipment (e.g., tower cranes, mobile cranes, etc., including accessories, such as slings), the collapse of machinery during use due to overloading or during erection and dismantling, failures due to poor slinging techniques, and other risks are associated with lifting equipment (e.g., tower cranes, mobile cranes, etc., including accessories, such as slings).
- Some of the actions that can be taken to limit the likelihood of accidents include demonstrating operator competence following machine-specific training, careful planning and supervision of the task, and efficient inspection, repair, and repair procedures.

Cranes

Cranes can cause significant and serious injuries, if they are not examined before use and are not used properly. These injuries frequently occur when a worker is injured by an overhead weight, or gets entangled in the crane's swing radius. Many crane accidents occur when the crane's boom or load line collides with an overhead electrical wire.

Guidelines

- Before using a crane, inspect all controls to guarantee good operation.
- Make certain that the load does not surpass the rated capacity of the crane.
- Raise the crane a few millimetres to ensure the brake system's balance and efficacy.
- Before using the crane, inspect all rigging. Do not wrap hoist ropes or chains around the cargo.
- Extend the outriggers completely.
- Never shift a load over workers.
- Block off all entry points inside the crane's swing radius.
- Keep an eye out for overhead electrical distribution and transmission lines, and keep a safe working distance of at least 1 foot from energised electrical lines.
- When working near energised transmitter towers, provide an electrical ground to the upper rotating support structure for the boom and materials being handled.
- The rated load capacity, operating speed, and directions are displayed and visible to the operator.
- Cranes have a load chart, and the operator may always calculate the angle and length of the crane boom.
- Crane equipment and other rigging machinery are examined daily before use to ensure that they are in good working order.
- Tag lines are used to prevent harmful swings or rotation of materials when a crane or derrick raises or lowers them.
- Illustrations of crane and derrick operator hand signals are exhibited prominently on the job site.
- The signal person gives the crane operator the correct signals.
- Anti-skid surfaces are used on crane platforms and walkways.
- Broken, worn out, or damaged wires and ropes must be replaced.
- Guard rails, handholds, and steps ensure safe and simple access to and from all sections of the crane.
- Only appropriately trained and qualified operators may use lifting and rigging equipment.



Fig. 1.2

Forklifts

- Forklift truck operators have proven their ability to safely operate these vehicles by completing training and evaluation.
- No worker aged less than 18 years is permitted to use a forklift.
- Forklifts are inspected daily to ensure that the brakes, horns, steering, forks, and tyres are in good working order.
- Written consent from the truck manufacturer is acquired for any alterations or additions that influence the vehicle's capacity or safe operation.
- In battery charging facilities, precautions are taken to prevent fires, sparks, or electric arcs.
- When a forklift operator is spotted operating the vehicle in a hazardous manner or when one is assigned to operate a different type of truck, refresher training and evaluation are provided.
- When a powered industrial vehicle is left unattended, the load and forks are fully lowered, the controls are neutralised, the power is turned off, and the brakes are set.
- There is enough elevation for the forklift and operator under the above installations, such as lighting, pipes, sprinkler systems, and so on.
- There are overhead protections to protect the operator from falling debris.
- Educate and license all operators to guarantee safe forklift operation, and maintain hauling equipment, particularly, tyres.

- Before utilising a forklift truck, inspect it for flaws.
- Use safe operating practices when picking up, moving, lowering, and stacking loads.
- Never drive faster than 5 mph and slow down in busy or slippery areas.
- Remove forklift vehicles that are hazardous or faulty from operations immediately.
- Disallow reckless driving.
- Do not transport loads that are heavier than the industrial truck's capacity.
- Drivers must always wear seatbelts.
- Avoid transporting heavy loads.
- Check if the rollover protecting structure is in place.
- Check that the reverse motion warning is operable and audible above the background noise.



Fig. 1.3

Electricity Related Risks

- Fires are caused by poor electric equipment and faulty electrical appliances. The most serious risks associated with electricity are electrocution and burns. Many deaths and injuries are caused by the following factors:
  - Use of poorly maintained electrical equipment
  - Working near overhead power lines and mains electricity cables
  - Contact with underground cables during excavation work or horizontal boring or drilling
  - Use of unsuitable electrical equipment in explosive areas
- Operating near high voltage power lines (including building systems, overhead power lines, and subterranean cables) can result in fatal accidents because of direct contact with live wires or lines arcing from surrounding equipment. Electrical accidents can occur at a wide variety of voltages, but the risk of harm is often greater with higher voltages. Electric shock and burns, loss of muscular control, thermal burns, and other injuries may be caused by alternating current (AC) and direct current (DC) electrical supply.

Trenching and Excavations

- Excavation and trenching are two of the most dangerous construction activities. An excavation is defined by OSHA as any man-made cut, cavity, trench, or dip in the Earth's surface caused by soil removal. A trench is an underground excavation that is more shallow than it is wide and is no more than 15 feet wide (4.5 m).
- Every year, collapsing of trenches kills people and injures many. Cave-ins represent the greatest risk and are far more likely to result in worker fatalities than other excavation-related mishaps.
- Falls, falling loads, toxic atmosphere, and mishaps involving mobile equipment are all possible risks.

Guidelines

- Never walk into an unguarded trench.
- For ditches that are 5 feet or deeper, always utilise a protective system.
- Hire an engineer to develop a trench protection system for ditches that are 2 feet or deeper.

System of Protection

- Sloping is done to safeguard personnel by drastically cutting the trench wall at an inclination away from the excavation that is

not steeper than the height/depth ratio of 1.5:1, based on the scoping criteria for the type of soil.

- Shoring is done to protect the workers by installing anchors to minimise soil movement in tunnels that are not deeper than 20 feet.
- Shielding is done to protect workers from soil cave-ins by putting in place trench boxes or other sort of supports.
- Always provide a means for workers to exit the trench, such as a ladder, stairs, or ramp, with not more than 25 feet of lateral motion.
- Place spoils at least two feet away from the trench's edge.
- Inspect trenches before entering them and after any hazard-encouraging occurrences, such as rainfall, tremors, or large surcharge loads.

Ladders

Ladders and stairways are also common causes of injuries and fatalities among construction workers.

Guidelines

- A competent person must check a ladder before use for defects, such as damage to the structure, split/bent side rails, damaged or missing rungs/steps/cleats, and missing or damaged safety devices.
- Ensure that ladders are free from grease, dirt, or other contaminants that may cause slips or falls.
- Make sure that the ladders are long enough to reach the work area safely. Mark or tag malfunctioning ladders for repair, or remove and replace them immediately.
- Never exceed the maximum intended load or the manufacturer's rated capacity when loading ladders.
- Check that the ladder is able to bear the user's weight, including materials and tools. Avoid using metal staircases with near electrical work and power lines.

Stairways

Hazards like slips, trips, and falls on stairwells are a leading cause of injury and death among construction workers.

Guidelines

- Stairway treads and walkways must be clear of dangerous objects, debris, and slippery materials.
- Check that the treads cover the whole step and landing. Stairways with four or more risers, or those above 3 inches must have a minimum of at least one railing.

Safety Checklist

The following checklist may assist you in taking precautions to avoid risks that lead to injuries, illnesses, and deaths. Always be careful and seek assistance, if you are suspicious of a threat.

Personal Protective Equipment (PPE)

Eye and Face Protection

- Safety glasses or face shields are used while working as foreign substances may enter the eye, such as during welding, grinding, sharpening, or nailing (or while working with concrete and/or hazardous chemicals, or when exposed to flying particles).
- Eye and face shields are chosen based on potential threats. When working on electrified electrical systems, wearing safety glasses or face masks is of utmost importance.

Foot Protection

- When operating around large machinery or falling objects, safety-toed footwear is provided to workers to prevent foot or toe injury.
- Workers must wear slip-resistant shoes with puncture-resistant soles.



Fig. 1.4

Head Protection

- A head injury might permanently disable a worker, or may even be fatal. One of the simplest methods to safeguard the head from injury is to wear a safety helmet. Workers can be protected by hard helmets from impact and penetration dangers, as well as, electrical shock and burns.



Fig. 1.5

- Employers must instruct the workers to wear helmets, if any of the following conditions exists.
  - Falling objects that may hit the head
  - Fixed objects, such as unprotected pipes or beams that may lead to collision
  - Unforeseen head contact with electrical hazards
- Hard helmets are an important personal protective equipment that workers at a construction site must always wear. Such hats must have a hard outer shell and a shock-absorbing liner with a headband and straps that hang the exterior from 1 to 1.25 inches (2.54 cm to 3.18 cm) away from the wearer's head. Such a hat absorbs shock in case of a collision and allows airflow during normal wear.
- There are three industrial classes of hard hats.
  - Impact and penetration resistance, as well as, limited voltage protection, are provided by Class A hard hats (up to 2,200 volt).
  - Class B hard helmets offer protection against electrical hazards, including high-voltage shock (up to 20,000 volt) and burns. They also defend the worker from flying/falling objects
  - Class C hard hats offer lightweight comfort and pressure protection but no electrical protection.
- A 'bump hat' is another type of head protection available in the market that is designed for usage in situations with little head clearance. They are suggested for regions where head knocks and lacerations must be avoided.
- Workers must wear hard hats when there is a risk of falling objects, or inadvertent head contact with electrical risks.
- Hard helmets are inspected regularly for dents, cracks, or degradation. After a violent blow or electric shock, hard hats are replaced.

Electrical Security

- Work on electrified (hot) electrical circuits, both new and old, is forbidden until power is disconnected and grounding is done.
- A reliable lockout/tagout system exists. Electrical cords or cables that are frayed, broken, or worn out are replaced as soon as possible.
- Grounding prongs are found on all extension cables. Guard against damage to flexible cords and cables. Sharp corners and extensions should be avoided at all times.
- Use three-wire extension cable sets with electrically powered tools and appliances designed for hard or extra-hard service (look for the letters S, ST, SO, and STO embossed on the casing.)
- All electrical tools need to be stored safely, checked for flaws regularly, and decommissioned immediately, if a defect is discovered.
- Do not circumvent any protection system or equipment meant to safeguard workers from electric contact.
- Identify and locate overhead electrical power lines. Keep ladders, scaffolds, equipment, and materials at least 10 feet away from power lines.
- Unless they are double insulated, all electrical instruments must be properly grounded. Multiple plug adapters must never be used at a construction site.

Floor and Wall Openings

- Floor openings (12 inches or more) are protected on all sides by a secured cover, guard rail, or equivalent (except at entrances to stairways).
- Toeboards are placed around the perimeter of permanent floor holes (where persons may pass below the opening).

Surface Elevations

- When applicable, signs displaying relatively high surface load capacity are installed.
- Standard guard rails are installed on surfaces that are raised more than 48 inches from the floor or ground.
- Standard 4-inch toeboards are installed on all elevated surfaces (below which humans or machinery may be exposed to falling debris).
- Elevated storage and work surfaces have a permanent method of entry and exit guarded by railings. The materials are stacked, piled, or racked in a way that prevents them from toppling, falling, collapsing, tumbling, or spreading.

Safety in Demolition

- Safety and health officers frequently have a bleak assignment in identifying and documenting the safety and health violations that lead to the maximum fatalities. The risks associated with demolition work, on the other hand, can be mitigated and eliminated with adequate planning, provision of personal protective equipment, adequate training, and adherence to the OSHA guidelines.
- Demolishing, razing, dismantling, or destroying any building structure, or any part of it is referred to as demolition. Many risks associated with buildings are included in demolition activities. However, demolition poses additional risks due to unknown circumstances, making the process hazardous. They are as follows.
    - Changes to the structure's design incorporated during construction
    - Approved or disapproved changes that affect the original design
    - Materials concealed inside structural elements, such as lead, asbestos, silica, other substances, and heavy metals, necessitating particular material handling
    - Unknown building material strengths and weaknesses, such as post-tensioned concrete
- So, everyone at a demolition site must be aware of the potential hazards and the safety measures that must be adopted. Specific guidelines for the construction sector to handle demolition hazards must be put in place.

Hazards Due to Demolition

- Planning is required to ensure that a demolition operation goes off without accident or injury. A competent person must conduct an engineering survey before a demolition activity is carried out. This should cover the structure's condition, as well as, the likelihood of an unintentional collapse.
- Before any demolition activity begins, a health risk assessment must also be performed. Besides these, planning must include the following.
    - A competent person must conduct an engineering survey before any demolition work begins. This should cover the structure's condition, as well as, the likelihood of an unintentional collapse.
    - Identifying, protecting, and/or relocating any adjacent utilities
    - A plan for fire prevention and evacuation
    - Emergency medical services and first aid
- Workers must be equipped with PPE kits while carrying out demolition activities. PPE helps the workers in the following ways.
    - Eye, facial, head, hand, and foot protection
    - Hearing protection
    - Respiratory protection
    - Added safety clothing (for example, cutting or welding operations)
    - Personal Fall Protection Systems (PFAS)
- However, mere provision of PPE is not enough. Workers must also be instructed in the selection, usage, fitting, inspection, maintenance, and storage of PPE.

- Most risks or threats that exist during demolition operations are the same as during other building tasks. However, there are additional risks associated with destruction due to a range of other reasons. Lead-based paint, sharps or protruding objects, and asbestos-containing material also pose a threat to workers.

Tips for Ensuring Safety during Demolition

- Reinforce or shore up the walls and flooring of damaged facilities into which personnel must enter.
- Before using PPE, inspect it.
- Choose, wear, and use task-specific PPE.
- Inspect and illuminate all stairwells, passages, and ladders.
- Turn off or close all electrical, gas, steam, sewer, and other service lines, and contact the concerned utility companies.
- Guard wall openings, up to 42 inches in height, cover and secure floor gaps with material capable of withstanding the weight that is expected to be imposed.
- Material disposal floor apertures shall not take more than 25 per cent of the entire floor space.
- Drop demolition debris to the ground or into debris bins using enclosed chutes with valves on the discharge end.
- Exterior wall and floor demolition must start at the top of the structure and work its way down.
- No structural or load-bearing construction on any floor may be altered or replaced until all storeys above that floor have been demolished.
- Before tearing down the walls, all roof cornices and other ornamental masonry must be removed.
- Employees must not be allowed to work where structural collapse hazards occur until they are rectified by shoring, bracing, or other efficient mechanisms.

Management of Hazardous Waste

- Hazardous waste management is the collection, treatment, and disposal of waste. If not handled appropriately, these can cause significant health and environmental hazards.
- Hazardous wastes might be solids, liquids, sludge, or confined gases. They are primarily produced as a result of chemical production, and other industrial activities. They can cause damage during improper storage, transit, treatment, or disposal.
- Surface and groundwater supplies are commonly contaminated by improper hazardous waste handling or disposal. People who live in houses built near old and defunct trash disposal sites may be especially exposed to such wastes.
- The government must strictly impose the hazardous waste management guidelines to address the present problems and prevent potential suffering because of hazardous wastes.

Characteristics of Hazardous Waste

- Hazardous wastes are classified on the basis of their biological, chemical, and physical properties. These properties result in poisonous, volatile, flammable, corrosive substances, and infectious or radioactive compounds.
- Toxic waste is dangerous even in the smallest or trace amount. They could have immediate ramifications. They may be lethal or cause severe illnesses, or may not have long-term consequences that result in permanent harm. Some of them are carcinogenic. This means that after several repeated exposure, they can cause cancer. Some of them are mutagenic, causing significant biological changes in exposed human and wildlife progeny.
- Chemically unstable volatile wastes react strongly when exposed to air or water. They cause explosions or dangerous fumes.
- Low ignitable waste may lead to fires.
- Corrosive wastes contain acidic or alkaline substances. They destroy solid matter and living tissues on contact with chemicals.
- Infectious garbage includes old bandages, syringes, other hospital goods or institutions of biomedical research.

Transporting Hazardous Waste

- Hazardous waste generated in an area frequently need to be transported to another authorised facility for treatment, storage, or

disposal (TSDF). Because of potential hazards to public health, transportation receives special consideration from the government.

- Apart from occasional unintentional spillovers, hazardous waste, particularly, liquids, are purposely spilled or deposited in random locations. This practice is known as midnight dumping. This has been greatly reduced by the transportation and monitoring of all hazardous wastes.

Vehicles of Transportation

- Maximum hazardous waste is typically transported by road. Only a small amount is transported by rail, and almost none via air or canal. Highway transport is the most popular as it allows cars to visit a majority of industrial sites and authorised TSDFs. Railroad trains require expensive siding facilities and are suitable only for large waste transfers.
- Tank trucks, with a capacity of up to 34,000 litres, made of steel or aluminium alloy can haul hazardous waste. They can also be packaged and shipped in 2-litre barrels. Specifications and regulations for cargo tank vehicles and shipping containers are included in government statutes.

The System of Manifestation

- The 'cradle-to-grave' manifest system, which tracks the movement of hazardous material, from origin to destination, is an important part of waste transport regulations in the United States. The manifest mechanism helps to eliminate the problem of night dumping. It also provides a method for verifying the quality and quantity of waste produced, as well as, emergency actions in the event of an unintended spill.
- A manifest is a piece of documentation required by a hazardous waste generator, such as the chemical industry. The generator is ultimately responsible for garbage disposal and must give the manifest, as well as, the trash to a licensed waste haulier. The transporter must give the trash recipient a copy of the manifest at an authorised TSDF. Each time the garbage is transported, a copy of the declaration must be signed. Each participant keeps a copy of the manifest, and extra copies are distributed to approved environmental organisations.
- If a hazardous waste leak or unintended spill occurs during transit, the carrier must take immediate and appropriate action, including notifying the concerned local authorities of the discharge. To contain the waste, an area may have to be diked, and procedures to remove the waste and reduce environmental or community health problems must be implemented.

Treatment, Storage, and Disposal

- There are various hazardous waste management alternatives. The best approach is to reduce waste at the source, or to recycle the resources for other purposes. While waste reduction and recycling are appealing ideas, they are not regarded as the final solution to address the problem of hazardous waste disposal.
- Some amount of hazardous waste will still need to be treated, stored, or disposed of.

Treatment

- Chemical, thermal, biological, and physical processes may be applied to treat hazardous waste.
- Chemical procedures include ion exchange, precipitation, oxidation or reduction, and neutralisation.
- High temperature incineration, which not only detoxifies but also destroys some organic wastes, is the thermal approach. A specialised thermal equipment is used to burn waste in solid, liquid, or sludge form. These include fluidised bed incinerators, multiple hearth furnaces, rotary kilns, and liquid injection incinerators. One of the difficulties encountered during hazardous waste incineration is the likelihood of air pollution.
- Some organic wastes, particularly those generated by the petroleum industry, may be subjected to biological treatment. One method of biologically handling hazardous waste is landfarming. In this method, waste is appropriately mixed with surface soil.

Microbes that break down waste and nutrients may be added. Genetically engineered bacteria are used in some cases. Food and grazing crops are not grown in the same field.

- Physical waste treatment collects, solidifies, or reduces waste volume. Physical processes include evaporation, sedimentation, flotation, and filtering. Solidification is another process that involves encasing waste in concrete, asphalt, or plastic. Encapsulation creates a solid mass of material that is resistant to leaching. Waste can also be blended with lime, fly ash, and water to produce a cement-like substance.

Surface Storage and Land Disposal

- Hazardous wastes that cannot be destroyed through burning or other chemical processes are eventually disposed of on land. However, this is not an acceptable strategy due to the inherent environmental risks involved. The two most frequent methods of land disposal are landfilling and underground injection. Surface storage or containment systems are commonly utilised as a stopgap solution before land disposal.
- A lagoon is a popular type of temporary storage facility for hazardous liquid waste. It is an open pit or a holding pond. New lagoons must be filled with impenetrable clay soil and permeable membrane liners to preserve groundwater. Between the liners, leachate collection systems and underground monitoring wells must be built. Other than minor deposition, evaporation of volatile organic compounds, and maybe, some surface aeration, open lagoons provide little waste treatment. Accumulated sludge must be removed regularly and treated as hazardous waste.
- Many older, unlined waste piles, and lagoons are located atop aquifers used for public water supply, posing significant health and environmental risks. Many of these old sites have been identified and are being cleaned or corrected.

Safe Landfills

- Hazardous solid or containerised garbage disposal is subject to tougher laws than municipal solid waste disposal. Hazardous wastes must be disposed in secure landfills with at least 3 metre (10 feet) gap between the landfill bottom and the groundwater table or bedrock. A secure hazardous waste dump requires two impermeable liners and leachate management technologies.
- The twin leachate collection system is made of perforated pipes put above each liner. The upper system keeps leachate from becoming trapped in the fill, while the lower system serves as a backup. Leachate is collected and sent to a treatment plant via a pipe. A finished landfill is covered with an impermeable cover or top to decrease the amount of leachate in the fill and danger of environmental damage.
- Another method for disposing liquid hazardous waste is deep-well injection. This technique includes pushing liquid waste through a porous material, such as limestone or sandstone with a steel shell. High pressures are utilised to force the liquid into the pores and fissures of the rock, where it will be held indefinitely. The injection zone must be beneath an impermeable layer of rock or clay, and cannot be deeper than 0.8 km (0.5 miles) beneath the surface. Deep well injection is often inexpensive and produces nearly no waste, but it risks leaking hazardous material, and eventually contaminating sub-surface water supplies.

Corrective Measures

- Hazardous trash dumped in inaccessible terrain, ponds, or lagoons endanger both human health and the ecosystem. Previously, many comparable uncontrolled disposal sites were used, and later abandoned. Depending on the level of risk, many locations may require remediation. In certain cases, the threat may warrant a rapid response. In other circumstances, engineering studies may be required to investigate the condition before taking corrective action.
- One method for clean-up is to remove all waste from the site and transport it to another location for treatment and disposal. Off-site solutions are usually the most expensive

option. On-site clean-up is a method that limits leachate formation and mitigates the risk of groundwater contamination. Temporary waste removal, establishment of a secure landfill on the same site, and appropriate waste replacement may all be part of the on-site clean-up process. Contaminated soil or groundwater clean-up may also be included. Treated soil can be replaced on-site, and treated groundwater can be returned to aquifers by deep well injection.

- Full waste confinement is a less expensive solution. This is performed by creating an impermeable cover well over the hazardous waste site and impeding groundwater lateral flow using sub-surface exclusion walls; cut-off walls can be utilised for this purpose, if there is a native layer of impervious soil or rock underlying the site. The walls that surround the building are built deep enough to pierce the impervious layer. They can be dug around the site as trenches without disturbing or transporting the waste. To prevent collapse during construction, the trenches are filled with a bentonite clay slurry, then replaced with a soil-cement combination that solidifies to form an impenetrable barrier. Cut-off walls operate as vertical barriers to flowing water, while the impervious layer works as a bottom barrier.

Electrical Safety

- Power is typically delivered at 230 volt (single-phase) and 4 volt (3 phase) at workplaces, while some bigger workplaces may receive electricity at a higher voltage. The following information applies to workplaces that use 230–400 volt power supplies.
- The main hazards in electrical safety are contact with live parts generating shock and bums, faults that could lead to fires, or explosions, where electricity could be the source of combustion in a potentially flammable or explosive setting, such as in a spray paint booth.
- The danger of injury from electricity is intimately tied to where and how it is utilised, with wet and/or damp environments posing a greater risk.

Fundamentals of Electricity Contact

- The impact of exposure to electricity is determined by the level of voltage to which the body is subjected and the obstacle to the flow of electrical current offered by the body.
- The degree of the effect of electric shock on the body is determined by the following factors.
  - Voltage level
  - Degree of body resistance to current flow
  - Current course through the body
  - Length of time the current travels through the body
- If a worker comes in contact with electricity, he or she may be unable to disconnect from the power source. The human body is an excellent electrical conductor. If you touch someone, who is in contact with an electrical source, electricity may flow through your body, generating electric shock. To begin with, try to switch off (disconnect) the electricity supply. If the electrical source cannot be shut off easily and securely, remove the individual from the electrical source using a non-conducting material, such as a fibre glass object or a wooden pole.
- It is the employer's responsibility to ensure that before utilising portable equipment, extension cables and other flexible leads, which are particularly susceptible to deterioration to plugs and sockets, and their connections are examined, maintained, and if required, replaced. The outer sheath of flexible cables should always be firmly fastened to prevent the wires (particularly, earthing) from slipping out of the terminals.
- When connecting cable lengths, use appropriate cable connectors or couplers. Taped joints are not permitted.
- Electrical installations must be maintained and checked regularly by a competent person.
- Socket outlets must not be overloaded by the use of adaptors.
- The provided electrical equipment must be fit for use.

- Fixed electrical equipment should have an identified switch to disconnect power in case of an emergency.
- Portable equipment labelled as double insulated has the live and neutral properly connected to the plug by a competent person unless the plug is damaged.
- When there is risk of explosive environment, a specialised electrical device is necessary.

Managing Risk

Lower the Voltage

- Portable equipment that is frequently available is powered by a 110-volt supply, via a simple transformer. It is frequently centre tapped to the Earth, limiting the maximum voltage between a live conductor and the ground to 55 volt.
- Battery-powered tools, such as drills and screwdrivers, can be used in place of mains-powered equipment.
- Lighting for temporary and handheld use can be supplied at 12, 25, 50, or 110 volt.

Check the Installation of Fuses

A fuse safeguards a gadget against overcurrent. When the current reaches its rated capacity, it is meant to blow and cut-off power supply. It is critical to use a suitable fuse for a device. In general, 3 amp fuses are utilised in equipment rated up to 700 watts (W). A 13 amp fuse is required for equipment rated at more than 700 W. Some devices, such as television, require a 5 amp fuse, while others, such as printer, require 10 amp fuses. Always follow the manufacturer's directions.

Earthed Devices

- Class 1 equipment relies on being linked to the Earth via a socket for safety. If any live part of an equipment comes in contact with the casing or earthed parts of the equipment, the fuse will blow and the power will instantly be cut-off. The Earth bond test determines if there is a secure link to the Earth while testing Class 1 equipment. The insulation test ensures that there are no pre-existing Earth fault.
- Class 2 double insulated equipment has the sign double square, signifying that it is double insulated and so does not have an Earth wire. The equipment must have both basic and additional insulation, each of which is good enough to avoid electric shock. All internally electrified components are housed within an insulated body, preventing any contact with live elements. Double-insulated appliances in the EU are all labelled with a sign of two squares, one inside another.

Provide at least One Residual Current Device

- A Residual Current Device (RCD) can provide additional safety if an electrical equipment of 230 volt or more is utilised. RCDs are additional protection devices that do not prevent electric shock, but can decrease the length of some shocks by allowing the power supply to be quickly disconnected when electrocution occurs. RCDs are required on all circuits connected with portable equipment and on certain other circuits where the danger of electricity is heightened by the presence of water. An RCD is a device that detects certain, but not all, electrical system defects and quickly shuts off power supply. The optimal location for an RCD is in the electrical panel or the socket outlet, as this ensures that the supply wires are always protected. If this is not practicable, an RCD-equipped plug or a plug-in RCD converter can be used to provide further safety.
- RCDs for human protection must have a rated tripping current (sensitivity) of less than 3 milliamperes (mA).
- An RCD is a crucial safety device. Never bypass it.
- If the RCD trips, it indicates a problem.
- If the RCD trips frequently and no defect can be identified in the system, contact its manufacturer.
- The RCD has a test button that should be used regularly to ensure that its operation is free and functional.



Fig. 1.6

Perform Preventive Maintenance

- To avoid hazards, all electrical equipment and installations should be kept in good working order.
- This should include a periodic visual inspection, and if necessary, a testing system. Most electrical dangers can be controlled by focusing on a simple, low-cost strategy of looking for obvious evidence of damage or problems.
- It is suggested that fixed installations must be inspected and tested regularly by a qualified person. The regularity of inspections and testing will be determined by the type of installation, frequency of usage, and environment. Users can assist by reporting any damage or fault they encounter.
- Ensure that everyone dealing with electricity is qualified for the job role. Even basic chores, such as connecting a plug, can be dangerous; make sure everyone knows what they are doing before they begin with the maintenance work.

Engineering Industry Safety

Risks Associated with Metalworking and Woodworking Equipment

High risk of injury is associated with metalworking and woodworking equipment. Despite enhanced protection in place, several workers are injured every year while operating these machines. The facts reveal that injuries sustained in these accidents are usually more severe than those sustained in typical machinery accidents, and a substantial percentage of the victims endure lifelong impairment. Personnel who deal with equipment must be aware of the consequences and adhere to safe work standards. Lacerations, amputations, severed fingers, and blindness are common injuries suffered by workers using woodworking equipment. Wood dust and finishing chemicals are hazardous to health, and workers can also develop skin and respiratory problems.
Accidents caused by machinery include the following.

- Loose-fitting clothes, hair, and jewellery getting entangled in moving parts
- Discharge of materials when a machine is switched on
- Unintentional starting of a machine
- Slipping or stumbling into unprotected nip
- Contact with sharp corners, such as a cutting blade
- Making changes while the equipment is operational
- Unauthorised use of machines
- Inadequate preventive maintenance

Motions

- Rotating: nip points, spinners, shaft ends, and couplings
- Reciprocating: moving back and forth, up and down
- Transverse: moving in a straight, straight path

Points to Remember

Nip points are caused by rotating machine elements like rotating gear, pulley, chain, sprocket between grinding wheel and tool rest, and between rotating and fixed parts, shafts, and couplings.

Operations

- Cutting tools: bandsaws, drills, milling machines, and lathes
- Punching machines: punch presses and notchers
- Shears: mechanical, pneumatic, or hydraulic
- Bending: press brakes, tube benders, and plate rolls

General Safety Guidelines

- Ensure that all equipment have effective and functional guards that seem to be in a location where machines are in use.
- Replace the guards as soon as possible following repair.
- Never lubricate, clean, adjust, or repair a machine while it is operating, instead stop the equipment and switch off the power.
- Even if the power is turned off, do not leave the equipment until it has completely stopped running.
- Always operate a machine under the supervision of an authorised person.
- Do not use your hand or body to stop the machine.
- Before starting a machine, ensure that the work and cutting instruments are properly clamped.
- Clean metal chips, curls, or waste pieces on the floor immediately, and place them in designated containers.
- Only one worker should handle a machine or switch when working with another.
- Do not rest on a machine.
- Concentrate on your task and avoid unnecessary talking when operating a machine.
- The worker handling a machine must not be disturbed at work.
- Do not operate a machine, if you are tired or unsuited for the job.
- If a mechanical feed is not possible, use pushing sticks or avoid putting hands near cutters or saw blades.
- Keep basic first aid kit at an accessible location.
- Make sure that the machines are kept in well lit areas.
- Keep all moving parts and cutting heads away from your hands.

Guidelines for Clothing and Safety Equipment

- When using any equipment, always use safety glasses or facial protection designed for the type of job.
- If substantial work is being done, do not forget wearing industry-specific protective boots like gumboots.
- When necessary, proper hearing protection, eye shields, and face masks should be worn.
- Avoid wearing loose-fitting clothes, rings, watches, bracelets, or other jewellery pieces that may get entangled in machine parts.
- Avoid wearing neckties or any loose-flowing garments.
- Dress in shirts or uppers with sleeves cuffed or rolled above the elbows.
- Always take off your gloves before starting or using a machine. If the substance is rough and sharp, gloves must be used, or the material must be handled with the machine switched off.

Housekeeping Guidelines

- Keep the floor clean of oil, grease, or any other liquid. Spilt liquids must be cleaned immediately as they may lead to slips, trips and falls.
- Always keep the aisles free to avoid tripping or other accidents.
- Store goods in a way that prevents them from becoming tripping hazards.
- Never leave any tool unattended, or work on a machine's table. Even if a machine is not in use, tools or labour may slide off and injure someone.
- After usage, keep any portable tool to its proper storage location.
- Remove cuttings using a vacuum, brush, or rake. As cuttings are sharp, do not handle them with bare hand.
- Before turning on the power, make sure that all keys and adjusting wrenches have been retrieved from the machine. Provide a handy tool rack.
- Keep the work space clean and well-lit. The flooring should be level and non-slippery.
- Make sure that there is enough space around the equipment for the task to be done safely.

Machine Guarding Principles

- The fundamental premise is to evaluate the effectiveness of machine guarding to eliminate or reduce danger. However, there are certain general guidelines.
- All protections must be properly designed and constructed.
- They should be strong, resilient, difficult to override, and securely mounted at a position where there is frequent access to a danger zone. It is generally preferable to line the guard with a device that separates the power supply and arrests risky machine movements when the guard is opened. This includes not only providing guards but also training, maintaining oversight, and delivering stiff punishment.

Classification of Guards

- Fixed guards: these are intended to prohibit access to harmful areas of a machine by having a direct barrier that prevents both purposeful and accidental access. If an aperture is required for a fixed guard to manually feed in the material, it must not give access to the danger zone. In most instances, this is accomplished by placing the opening at a suitable distance from harmful machine elements.
- Interlocked guards: when this sort of guard is removed or withdrawn, the tripping mechanism and/or power shut down(s) or disengage(s), and the machine cannot cycle or start until the guard is replaced. An interlocking guard can be powered by electrical, mechanical, hydraulic, or pneumatic means, or by a combination of any of these. If 'inching' by remote control is required, interlocks should not be disabled. Changing the guard should not cause the machine to restart. To prevent occupational dangers, all movable guards should be linked.
- Adjustable guards: these are mobile guards that are modifiable for specific procedures and are generally fixed while in use. Some guards require the assistance of an operator to be placed into position; these guards must always be installed on the machine, and then, correctly placed and used every time. Self-adjusting guards are employed in some circumstances; however, they only give a partial solution to the challenges associated with manually adjustable guards. When maintained, self-adjusting guards are usually better than ones that require routine manual adjustment. As there will always be some accessibility to harmful areas, operators of machines with changeable guards must be adequately taught and supervised. They must be instructed on how to utilise the guards correctly, and they must exhibit competence and dependability. Finally, adjustable guards should be used in conjunction with additional safety equipment, such as push sticks, chain mail gloves, components jigs, guides, or vices.
- Self-adjusting guards: the movement of the stock determines the apertures of these barriers. As the operator pulls the stock into the danger zone, the guard is pushed away, leaving only a small opening to allow the stock. The guard restores to its rest position after the stock is withdrawn. This guard shields the operator by putting a barrier between the danger zone and the operator. The guards could be made of plastic, metal, or other strong material. Self-adjusting guards provide varying degrees of protection.
- Trip devices: these are intended to bring a machine to halt and guarantee safety before or as a person reaches the danger zone. Sensitive screens and barriers, telescoping arms, photoelectric safeguards, tripwires, and pressure-sensitive mats are a few examples of trip devices. They should have a reset mechanism to prevent the machine from restarting before the tripping mechanism is disengaged and the machine controls are used.

Welding and Gas Cutting Safety

- Oxy-fuel welding (also known as oxyacetylene welding, oxy welding, or gas welding) is a versatile technique for welding, cutting, heating, straightening, and cleaning. The apparatus is adaptable, inexpensive, and portable. It is so extensively used that individuals frequently

overlook the risks. Every year, many people are wounded as a result of erroneous or reckless usage of oxy-fuel gas equipment. Some injuries can even result in death.

- The main hazards are as follows.
  - Fire caused by heat, sparks, molten metal, or direct engagement with the flame
  - Explosion when cutting or rebuilding tanks or drums containing or potentially containing flammable material
  - Fire or explosion caused by gas leak backfire and flashes
  - Fumes produced during flame cutting
  - Fire or burns caused by the misuse of oxygen; burns caused by contact with flame or hot metal
  - Crushing or impact injuries sustained when handling and moving cylinders
- An oxy-fuel gas torch flame (sometimes known as a 'blowpipe') is a particularly intense source of ignition. Many fires have been started by careless usage of oxy-fuel torches. Combustible materials, such as wood, paper, board, fabric, rubber, and polymers catch fire easily. Many processes produce sparks and hot splatter, both of which can ignite these materials.

Fire Prevention Measures

The following measures will aid in fire prevention:

- Relocate the workpiece to a safe area for hot work.
- Remove surrounding combustible materials (such as flammable liquids, wood, paper, fabric, packaging, or plastic).
- Protect any nearby combustible objects that cannot be relocated. Make use of appropriate guards or covers, such as sheet steel, mineral fibreboards, or fire-resistant blankets.
- Make sure there are no combustible items hidden behind walls or in partitions, especially if welding or cutting is to be done for an extended period. Combustible insulation materials are used in some wall panels.
- Utilise flame-resistant sheets or covers to avoid hot particles from moving through openings in floors and walls (doorways, windows, cable runs, and so on).
- If the consequences of a fire are severe, such as working inside ships, you may need to assign a fire watch during and after the work is completed.
- Keep a fire watch for 30 minutes after hot work is completed.
- Keep flames, heat, sparks, or hot spatter away from hoses.
- Keep fire extinguishers handy.

Occupational Health and Safety in India

- With a projected annual GDP growth of 7–8 per cent, India is one of the world's fastest growing economies, fuelled by a massive expansion in sectors, such as chemicals, manufacturing, and textiles. Even though India has had occupational health and safety legislations in place for years, substantial segments of the estimated 465 million-strong workforce are not covered by the health and safety laws, and safety checks and enforcements remain small-scale and fragmented. As a result, work-related injury and illnesses continue to be major issues.
- The Constitution of India contains provisions on workers' occupational health and safety, including a responsibility cast on the government to focus its policies towards ensuring that the health and strength of employees, men, and women, as well as, children, is not exploited. Despite this, the health and safety legislations only cover four industries: factories, mining, ports, and construction. In many circumstances, these restrictions only apply to certain type of workplaces.
- The Factories Act, 1948, which is closely drafted on the basis and standards of the UK's Health and Safety at Work Act 1974, is India's primary health and safety legislation. In essence, it imposes a general duty on factory owners to ensure the health, safety, and welfare of their employees through measures, such as the delivery of safe plant and systems of work, as well as, appropriate data, instruction, training, and supervision. It also specifies tasks, such as factory registration, maximum hours worked, working temperature, machinery guarding, fire

prevention, and sanitation. Furthermore, the Act forbids employment of children below the age of 14 years, compels factories with more than 1000 employees to hire internal safety officers, and requires enterprises with more than 500 employees to appoint welfare officers.

- Local factory inspectorates in India's 28 states implement the Act, and they also have the authority to impose extra requirements under the statute.
- The Dock Workers (Safety, Health, and Welfare) Act of 1986 addresses the health, security, and welfare aspects of cargo loading and unloading, and is enforced by dock safety inspectorates in India's 11 major ports.
- The Mines Act of 1952 includes regulations for ensuring the health, safety, and welfare of coal, metal, and oil mines.
- The Child Labour (Prohibition and Regulation) Act of 1986 prohibits the employment of children below the age of 14 years in some dangerous processes, such as construction labour, etc.
- The Plantation Labour Act of 1951 stipulates the supply of welfare amenities and arrangements on plantations of five hectares or more employing 15 people, such as healthcare facilities, drinking water, sanitation, maximum hours worked, and rest intervals.
- Furthermore, the Indian construction industry is governed by health and safety regulations. The Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act 1996, in particular, establishes health, safety, and welfare regulations for construction enterprises employing 10 or more workers on projects costing more than ₹10 lakh. This includes the necessity to register the construction company with the appropriate authorities, designate a safety officer, and form a safety committee when 500 or more people are employed. Violations of the Act's safety regulations can result in fines, three to six months in prison, or both.
- Meanwhile, the Building and Other Construction Workers Welfare Cess Act of 1996 authorises state governments and union territories to levy 1 per cent cess on construction projects costing more than ₹10 lakh to fund state welfare schemes that provide financial assistance to workers and their families in the event of an accident.
- A variety of regulatory bodies are involved in the overall plan for ensuring occupational safety and health, and its implementation. The claims of pressure groups, however, diminish the effectiveness of the law and make it difficult to coordinate efforts in this area. The central government and the Ministry of Labour, in particular, are in charge of developing national policy and legislation, while the labour departments of India's 28 states and eight union territories are in charge of implementing and enforcing the Factories Act through their factory inspectorates.
- Meanwhile, the Directorate General, Factory Advice Services and Labour Institutes (DGFASLI), a central government agency, advises on the formulation of national health and safety policies for factories and docks, and liaises with state factory inspectorates on the Factories Act's implementation. The DGFASLI also monitors health and safety rules in India's 11 major ports through its docks safety Inspectorates.
- Meanwhile, the Directorate General of Mines Safety is in charge of ensuring mine health and safety, while state governments are in charge of enforcing the Plantation Labour Act 1951.
- Some of India's biggest corporations, particularly, international conglomerates, such as Tata, are promoting broader health and safety improvements via their actions. Many large organisations are at the frontline of corporate social responsibility and have a plan to constantly enhance health and safety. Some are looking beyond regulatory compliance and trying to promote their own health and safety agenda. Whatever the picture may be, it is apparent that much work remains to be done if India's spectacular economic growth is to be matched by comparable improvements in occupational safety and health performance.

Injury and Illness Compensation

- There is also legislation in India that allows select workers and their families to seek

compensation for work-related injuries and illnesses. Employers in control of factories, mines, plantations, construction work, mechanically propelled vehicles, and certain other hazardous occupations, in particular, must pay compensation to employees and their dependents in the event of work-related fatalities and serious injuries, as well as, certain occupational accidents, under the Employees Compensation Act 1923 (previously the Workmen's Compensation Act 1923). The Act currently allows for a maximum payment of ₹ 10.97 lakh.

- Meanwhile, the Employees State Insurance (ESI) Act of 1948 is a state welfare scheme that provides healthcare and cash benefits to employees.
- Factories that use authority and employ 10 or more people in the event of an occupational injury (including certain occupational diseases), illness, or maternity. The ESI programme, which is funded by employer and employee contributions, offers medical care for injured workers, as well as, annuities to the relatives of those who die as a result of work-related accidents or illnesses.

Factories Act, 1948

- The Factories Act of 1948 was drafted to consolidate and amend factory labour laws. It was enacted in April 1949 and applies to the entire country.
- As per Section 2 (m) of the Act, a factory means any premises, including precincts thereof:
  - where 10 or more workers work or were working on any day of the prior 12 months, in any part of which a production process is being carried on with the engine running or is ordinarily so carried on, or
  - where 20 or more workers are working or were working on any day of which a production process is being conducted without the aid of power or is ordinarily so carried on.
- Following the December 1984 Bhopal gas tragedy, a new chapter (Chapter IV-A) on 'provisions relating to hazardous processes' was added to the Act in 1987. The new rules address the formation of Site Appraisal Committees, the Occupier's duty for hazardous processes, emergency requirements, and allowed limits of exposure to chemicals and poisonous substances. Workers are also encouraged to participate in safety management. Some of the most essential aspects of the Factories Act's chapter 'Health and Safety, are described below.

Health

Section 11: Cleanliness

Every factory shall be kept clean and free of effluvial coming from any drain, privy, or other nuisance, particularly, dirt.

Section 12: Waste and Effluent Disposal

Every factory shall make effective provisions for the treatment and disposal of waste and effluent, resulting from the industrial processes being carried out therein.

Section 13: Temperature and Ventilation

Effective and appropriate facilities shall be made in every manufacturing workroom for ensuring adequate ventilation by the circulation of fresh air and preserving a temperature that will afford reasonable comfort to the workers and prevent injury to health.

Section 14: Dust and Fume

In every factory, as a result of the manufacturing processes, there is dust, fume, or other pollutants. They are likely to cause injury to workers. Effective precautions must be made to avoid inhalation and build-up in any workroom.

Section 15: Artificial Humidification

In respect of all factories, in which the humidity of the air is artificially increased, the state government may make rules prescribing standards of humidification; regulating the methods used for artificially increasing the humidity of the air; directing prescribed tests for determining the humidity of the air to be correctly carried out and recorded; and prescribing methods to be adopted for securing adequate ventilation and cooling of the air in workrooms.

- In any factory where the humidity of the air is deliberately raised, the water used for the purpose must come from a public supply or another supply of drinking water, or it must be adequately purified before use.

Section 16: Overcrowding
No room in any factory must be overcrowded to the point that it is harmful to the health of the workers employed there.

Section 17: Lighting
Sufficient and acceptable illumination, natural, or artificial, or both, shall be provided and maintained in any portion of a factory where workers are working or passing.

Section 18: Drinking Water
Every factory shall take appropriate measures to provide and keep a sufficient supply of healthy drinking water at suitable sites easily located for all personnel engaged within.

Section 19: Latrines and Urinals
Sufficient latrines and urinal facilities of prescribed types shall be provided in every factory at suitable and accessible places for workers at all times while they are at the factory.

Section 20: Spittoons
Every factory must have a substantial number of spittoons at convenient locations, and they must be kept clean and sanitary.

Safety

Section 21: Fencing of Machinery
The prime mover, flywheel, or any component of the stock bar that projects further than the headstock of a lathe, every component of an electric generator, transmission machinery, and any dangerous part of machinery shall be securely guarded by suitable guards of solid construction.

Section 22: Working on or near Machinery in Motion
If it is necessary for any factory to examine, or perform any adjustment, or lubricate, or relocate the belt of any part of a machine in motion, such inspection or operation shall be performed only by a specially qualified adult male worker wearing tight-fitting garments. The operator must have a secure foothold and handhold.

Section 23: Employment of Young People on Dangerous Machines
No young person shall operate on machines unless he has obtained suitable training in machine operation or is under the supervision of a person who has extensive knowledge and expertise with the machine.

Section 26: New Machinery Casing
Every setscrew, bolt, or key on any rotating shaft, spindle wheel, or pinion must be effectively guarded. Similarly, any spur, worm, and other gear that do not need to be adjusted frequently must be encased or covered.

Section 28: Hoists and Lifts
Every hoist and lift shall be of excellent mechanical construction and shall be closely investigated by a qualified person at least once every six months, and a register containing the particulars of each examination shall be maintained. Every hoist and lift shall be marked with the maximum safe operating load, and no load larger than that load shall be transported.

Section 29: Lifting Equipment
All sections of the lifting equipment, as well as, every chain, rope, and lifting tackle, must be of good construction. They must be maintained and examined by a qualified person at least once in every 12 months, and a register, including details of each examination, must be maintained.

Section 31: Pressure Plant
If any part of the plant or machinery utilised in the production process in any factory is operated at a pressure greater than atmospheric pressure, effective measures must be implemented to guarantee that the safe operating pressure of such part is not exceeded.

Section 32: Floors, Stairs, and Means of Access
All floors, steps, stairs, and corridors shall be of sound construction and properly maintained; and shall be kept free of obstructions and

substances likely to cause persons to slip; and where necessary to provide safety, steps, stairs, tunnels, and gangways shall be equipped with handrails.

Section 33: Pits, Sumps, and Floor Openings
All pits, sumps, and floor openings must be securely covered or enclosed.

Section 34: Excessive Weights
No one shall be engaged in the factory to carry, hoist, or move any extra load that may injure him/her.

Section 35: Protection of the Eyes
Effective screens or suitable goggles shall be provided for protection of the eyes of the person engaged in a processor near the process that involves:

- chance of harm to the eyes from particles or fragments thrown off in the course of the process
- risk to the eyes due to excessive light exposure

Section 36: Precaution against Dangerous Fumes
No one shall enter any tank, vessel, pipe, or confined space in a factory, where toxic vapours are expected to be present, unless another tank and vessel is supplied with adequate sized manholes or other effective means of exit. Workers must also wear appropriate breathing apparatus and a belt securely tied to a rope, the end of which is held by someone outside the confined space.

Section 36A: Precautions for the Use of Portable Electric Lights
No portable electric light or another device with a voltage more than 24 volt shall be used within any chamber, tank, pipe, or other confined place.

Section 40B: Safety Officers
In every factory where 1000 or more workers are normally employed, or any manufacturing process or operation is carried out that involves any danger of bodily injury, poisoning or disease, or any other danger to the health of the persons employed in the factory, the prescribed number of safety officers shall be employed.

Section 87: Dangerous Operations
Rules requiring measures to be maintained during the process or operation shall be made for any process or activity carried out in a factory that exposes employees to a serious risk of bodily damage, poisoning, or disease.

Section 88: Notice of Certain Accidents
When an accident occurs in a factory that causes death or bodily injury and prevents the person injured from working for 48 hours or more immediately following the mishap, the factory manager shall send a notification to the authorities in the form and within the time prescribed by the State Government.

Section 88A: Notification of Certain Harmful Occurrences
In any factory, if a dangerous event occurs, whether or not it causes bodily injury or disability, the factory management shall notify the authorities. Work at height and entry to constrained spaces are restricted by work permits and specified procedures.

Accident Causation Theories

Over the years, several theories of accident causation have evolved that explain the accidents that occur, but so far none has been universally accepted. Researchers from different fields of science and engineering have been trying to develop a theory of accident causation, which will help to identify, isolate, and ultimately remove the factors that contribute to or cause accidents. Various accident causation theories are as follows.

- Domino Theory
- Human Factors Theory
- Epidemiological Theory
- Systems Theory
- Combination Theory

Domino Theory

- Herbert W Heinrich is the proponent of his 'Axioms of Industrial Safety' and theory of

accident causation, which came to be known as the Domino Theory. Heinrich's model, known as Domino Model, was introduced in 1931.

- The Domino Model describes an accident sequence as a five-step series of events that occur in a fixed and logical order. According to the premise of the model, a set of unsafe conditions is similar to a row of vulnerable dominos.

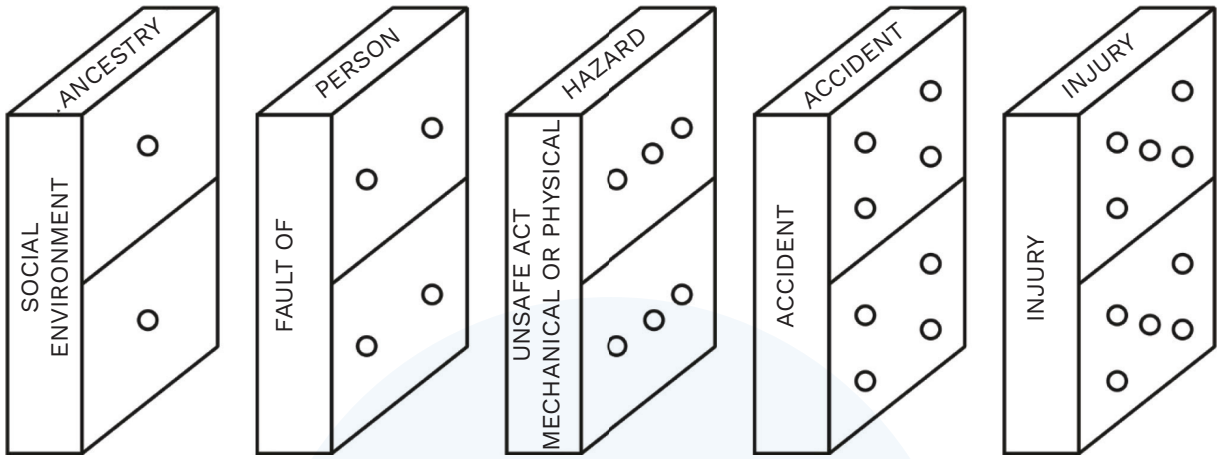


Fig. 1.7

- The model takes the form of five domino bricks in a row, representing five factors in the sequence leading to an accident. Chronologically, these factors can be summarised as follows:
  - Ancestry and social environment: people inherit (ancestry) or learn through the socialisation process certain behavioural characteristics, which are negative character traits (such as stubbornness recklessness, etc.) that might predispose them to behave in an unsafe manner.
  - Fault of person: the inherited or acquired negative character traits of people (such as recklessness, ignorance of safe work practices, violent temper, etc.) make people behave in an unsafe manner and that is how hazardous conditions exist.
  - Unsafe act, or mechanical or physical hazard: the direct causes of accidents lie in unsafe acts (such as standing under suspended loads, with removal of safeguards, horseplay, etc.) committed by people and the existing mechanical or physical hazards (such as unguarded gears, absence of rail guards, insufficient lighting etc.)
  - Accidents: events, such as fall of person(s), the impact of moving objects on people, etc., are typical accidents that result in injury.
  - Injuries: these include injuries that result directly from accidents, such as lacerations and fractures.

Human Factors Theory

- Human factors models of accident causation refer to distracting influences upon the worker by either internal or external factors. Such distracting influences are of a temporary rather than permanent nature. Therefore, if care is taken to eliminate such distracting factors, there is a possibility of preventing the causation of accidents.
- Among various human factors models, Ferrell's Human Error Theory and the Petersen Accident-Incident Causation Model will be discussed here.
  - Ferrell's human error theory: Russell Ferrell, professor of human factors at the University of Arizona, is the proponent of this theory. According to this theory, accident causation is attributed to a chain

of events ultimately caused by human error. Human error is caused by one of the three situations: overload, inappropriate response, and inappropriate activities.

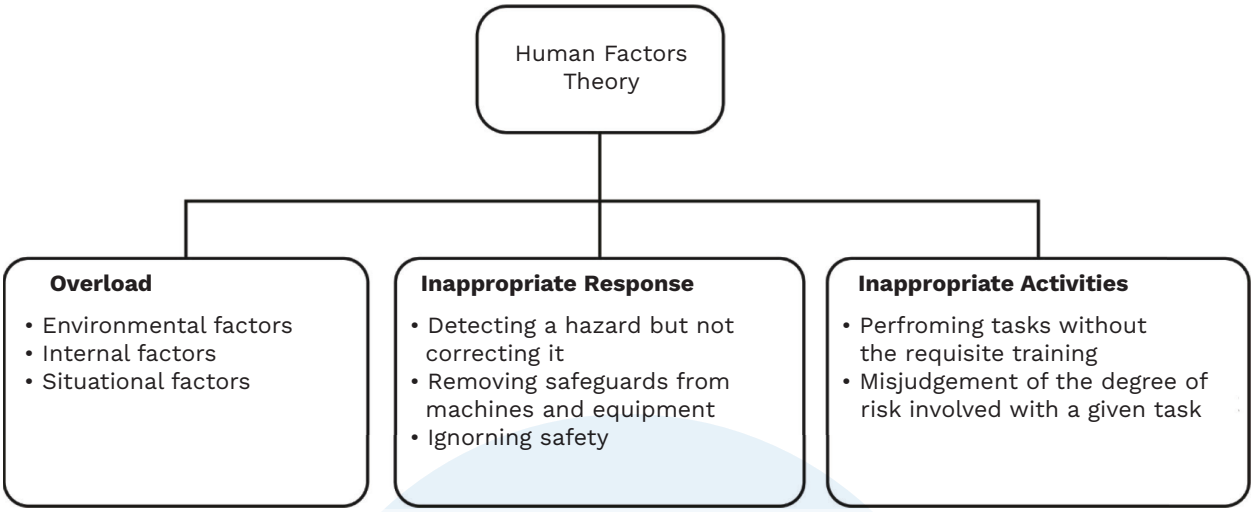


Fig. 1.8

- Overload: it amounts to imbalance between a person's capacity at a given time and the load that person is carrying in a given state. A person's capacity is the product of factors, such as his or her natural ability, training, state of mind, fatigue, stress, and physical condition. The load a person is carrying consists of tasks for which he or she is responsible and added burden resulting from environmental factors (such as noise, heat, cold, vibration, etc); internal factors (personal problems, emotional stress, anxiety, etc.) and situational factors (e.g., level of risk, unclear instructions, etc.)
- Inappropriate response/incompatibility: how an individual responds to a given situation can cause or prevent an accident from occurring. If a person removes a machine guard from a machine to increase output, he or she responds inappropriately. If a person disregards an established safety procedure, or he or she has responded inappropriately, then such responses can lead to accidents. In addition to inappropriate responses, this component includes workstation incompatibility. The incompatibility of a person's workstation regarding size, force, reach, feelings, and similar factors can lead to accidents and injuries.
- Inappropriate activities: human error can be the result of inappropriate activities, for example, a person undertakes a task but does not know how to do it, or a person misjudges the degree of risk involved in a given task, and tries to carry out the job based on that misjudgement. Such inappropriate activities may lead to accidents and injuries.

- Petersen accident/incident theory: in 1982, Dan Petersen, a safety consultant, proposed this model in his book titled *Human Error Reduction*. This model suggests that human error is caused by three broad factors, namely overload, ergonomic traps, and decision to err.

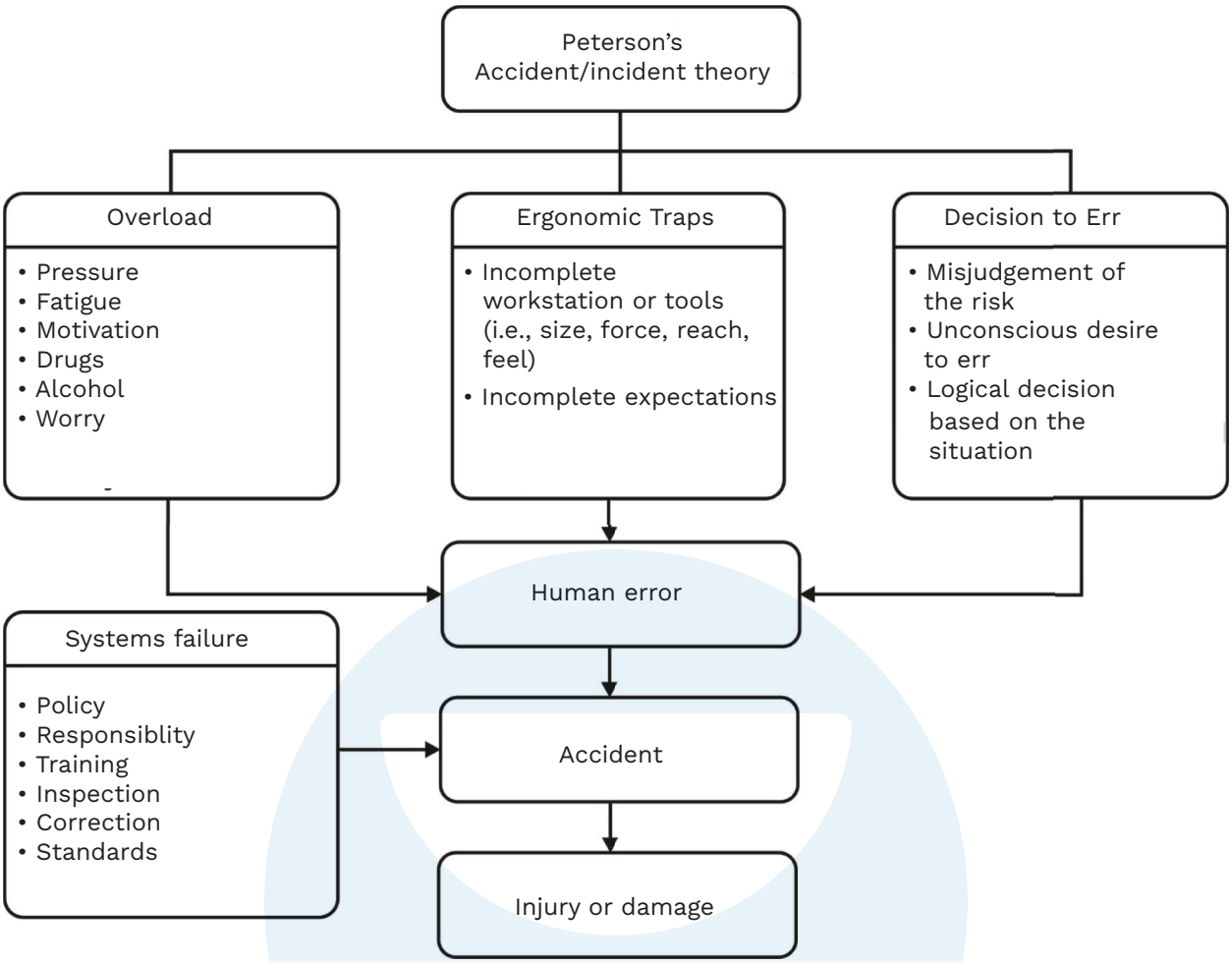


Fig. 1.9

- Overload is a mismatch of capacity with load in state. An individual's capacity depends on that person's natural endowment, physical condition, state of mind, knowledge, and skill, the habit of indulging in drug or alcohol abuse, fatigue or pressure, motivational state, attitudinal attributes, arousal level, and biorhythm. Load arising out of the tasks one has to perform challenges ones capacity. In the event of any mismatch between the load and capacity, there is a possibility of overload. Tasks may include information processing, environmental demands, worry and stress arising out of life change events (measured in terms of Life Change Units), situational hazards, etc.
- The decision to err suggests that often employees are found to commit human error as they take certain decisions consciously or unconsciously. There are many occasions when workers choose to perform a task unsafely. Such things happen due to peer pressure, expectation for appraisal, priorities of the management, and personal values. It may also happen as they perceive low probability of an accident happening to them, or because they perceive a low potential cost to them in case of an accident.
- Ergonomic traps occur due to faulty workstation designs and incompatible displays or controls. Thus, human error is the outcome of overload, decision to err, and ergonomic trap. In the causation of accidents, human error and system failure play a major role. Many potential system failures are related to

organisational factors. The management fails to establish a comprehensive safety policy. Responsibility and authority regarding safety are not clearly defined. Safety procedures, such as measurement, inspection, correction, and investigation are ignored or given insufficient attention. Employees do not receive a proper orientation. Employees are not given sufficient safety training. These are some examples of many type of system failures that might occur according to Petersen's Accident-Incident Causation Model.

Epidemiological Theory

- This theoretical framework in general terms explains the causal association between diseases, or other biologic processes (accidents) and specific environmental experiences.
- 'Heinrich's first three dominos would be host (human), agent (hazard), and (environment). All kinds of illness, including injury, are because of interactions between these three categories. The science of epidemiology was developed from the perspective of infectious diseases, which is also reflected in its vocabulary. Diseases were generally seen as a result of impacts from external agents, such as bacteria or viruses, affecting the host and environmental conditions that could either convey or restrain this process. From the perspective of the epidemiological model, an accident is defined as 'the unexpected, unavoidable unintentional act, resulting from the interaction of host, agent, and environmental factors within situations which involve risk-taking and perception of danger'.
- In this model, two important components, namely predisposition and situational characteristics, are instrumental in accident conditions and effects. Accident effects are injuries and damages inflicted on people or property. These effects are the measurable indices of an accident. Conditions under which an accident takes place are unexpected, unavoidable, and unintentional acts resulting from predisposition and situational characteristics. Predisposition characteristics include the susceptibility of people (host), hazardous environment, injury inflicting agents, etc. Situational characteristics are risk assessment by the individual, peer pressure, priorities of the supervisor, and a prevailing attitude. For example, if an employee who is particularly susceptible to pressure from

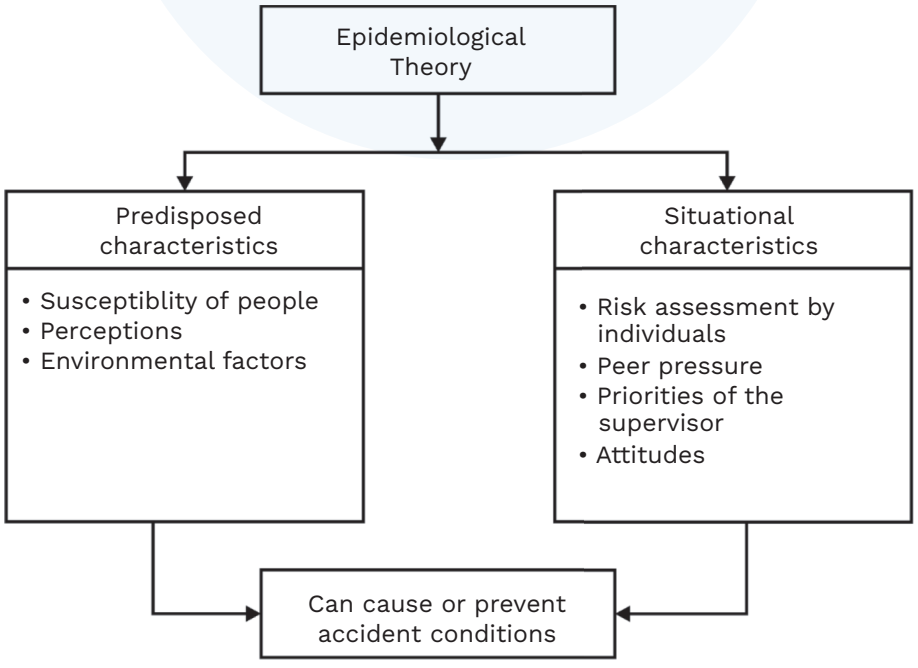


Fig. 1.10

people in position of power (predisposition characteristic) is pressured by his or her supervisor (situational characteristic) to speed up his or her operation, the result would be an increased probability of an accident.

Systems Theory

- The concept of 'systems' provides a general framework for modelling mutual and complex interactions in virtually all type of applications, from technology and biology to economy, psychology, and sociology. Using the systems theory, it is possible to describe the dynamics of such circumstances in more detail, and to understand under which circumstances a given system transforms into new, and perhaps, unwanted stages or modes of operation. In the late 1960s, Surry explored the nature of interaction from a behaviour and systems oriented view. This was done to understand why such interactions give rise to accident risks, and why risks are transformed into accidents and injuries. Surry's model consists of two sequences. The first analyses the risk build-up from normal man-machine-environment interaction, and the second describes the dynamics of accidents and what makes some result in injuries.

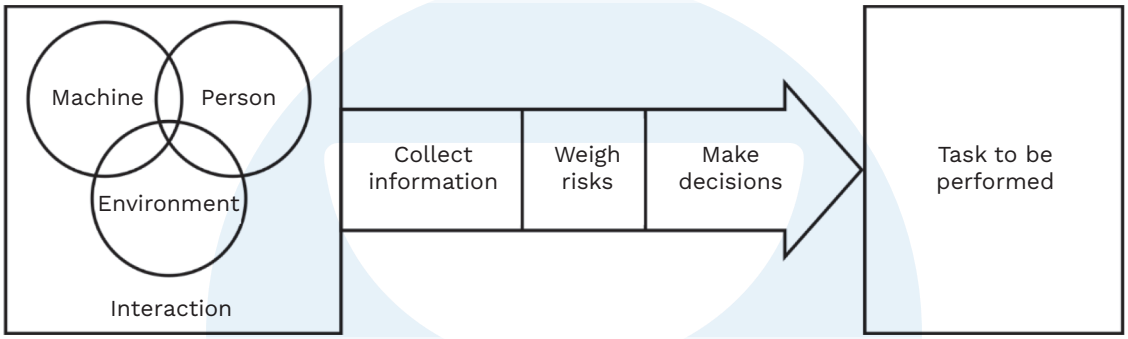


Fig. 1.11

- The most widely used systems model was developed by RJ Firenzie. The model has imported blocks. The first block consists of interaction among man-machine-environment. The second block comprises the process of collection of information, weighing of risks, and decision-making. Tasks under the third block are to be performed as the outcome of the processes involved in the previous two blocks. The feedback loop connects the third and the first block. It is evident from this model that as the person interacts with a machine within an environment to perform a task, three activities, namely collection of information, weighing of risks and decision-making to perform the task take place. Based on the information that has been collected by observing and mentally noting the current circumstances, the person weighs the risks and decides whether or not to perform the task under existing circumstances. For example, a machine operator is working on a rush order against the last approaching deadline. An important safety device has malfunctioned in the machine. Simply taking it off will interrupt work for only five minutes, but it will also increase the probability of an accident. However, replacing it could take up to an hour. Should the operator remove the safety guard and proceed with the task or take the time to replace it? The operator and his supervisor might assess the situation (collect information), weigh the risks, and make a decision to perform the task. For this reason, Firenzie recommends the five factors to be considered before beginning the process of collecting information, weighing risks, and making a decision. These factors are as follows:
  - Job requirements
  - The worker's abilities and limitations
  - The gain, if the task is accomplished
  - The loss, if the task is attempted but fails
  - The loss, if the task is not attempted

- It is, particularly, important to consider these factors when stressors, such as noise time, constraints, or pressure from a supervisor might tend to cloud one's judgment.

Combination Theory

- There is often a degree of difference between any theory of accident causation and reality. The various models presented with their corresponding theories in this chapter explain why accidents occur. For some accidents, a given model may be accurate. For others, it may be less so. Often the cause of an accident cannot be adequately explained by only one model or theory. Thus, according to the combination theory, the actual cause may combine parts of several different models. Safety personnel should use these theories as appropriate for both accident prevention and accident investigation. However, they should avoid the tendency to try to apply one model to all accidents.

Practice Exercise

1. Consider the following statements.
I. The failure of many structures.
II. Accidents on major highways.
III. Discharge of effluents which pollute rivers.
IV. Dangerous gas emissions.
These are attributable to the following:
1. Improper design
2. Not adhering to proper maintenance practices
3. Lack of proper inspection and quality control during construction
4. Corruption amongst several concerned individuals
5. Lack of public concern
Select the most appropriate answer using the codes given below.
(A) 1, 2, and 3 only
(B) 1, 2, and 4 only
(C) 3, 4, and 5 only
(D) 1, 2, 3, 4, and 5

2. Consider the following statements regarding safety during the demolition of a multistorey building.
I. Demolition need not proceed storey by storey.
II. Floor openings, not used for material chutes, should be enclosed with adequate guard rails.
III. Before demolition starts, all lath and loose plaster shall be stripped off throughout the building.
IV. Adequate and well-stiffened lateral bracing shall be provided for walls.
Which of the above statements are correct?
(A) I, II, and III only
(B) I and IV only
(C) II, III, and IV only
(D) I, II, III, and IV

3. In a health and safety context, a hazard is:
(A) anything that could result in a physical injury.
(B) the likelihood of someone being injured at the workplace.
(C) anything that could result in a psychological injury.
(D) anything with the potential to result in an injury or illness.

4. Risk management involves:
(A) hiring an employee with the job of managing risks.
(B) identifying and monitoring risks.
(C) eliminating all workplace hazards.
(D) identifying, assessing, and eliminating or controlling risks.

5. Workplace health and safety legislation imposes a duty of care on:
(A) the employer.
(B) the employee.
(C) all participants at the workplace.
(D) visitors to the workplace.

6. The Health and Safety at Work Act provides a comprehensive legal framework to:
(A) promote and encourage high standards of health and safety at the workplace.

(B) promote standards of responsibility at the workplace.
(C) encourage employers to be more responsible.
(D) encourage employees to be more responsible.

7. Which of the following fire extinguishers should not be used to extinguish a liquid fire?
(A) Water
(B) Powder
(C) Carbon dioxide
(D) Foam

8. What should an employer do as part of health and safety support for staff?
(A) Issue staff with a written contract of their employment.
(B) Undertake regular customer surveys to ascertain the level of satisfaction with the services provided.
(C) Ensure that all staff is aware of the safety procedures, by providing safety information and training.
(D) Make sure that there is a one-to-three staff to supervisor ratio.

9. The two most important factors in creating a culture of safe work practices are:
(A) praise and discipline.
(B) networking and compliance.
(C) management commitment and employee involvement.
(D) decision-making and problem-solving.

10. Which of the following is not an element of the fire triangle?
(A) Oxygen
(B) Heat
(C) Fuel
(D) Flame

11. Which of the following is not a measure to control electrical risk?
(A) Earthing the equipment
(B) Providing RCD
(C) Machine guarding
(D) Reducing the voltage

12. Statement I: one benefit of implementing an effective safety and health management system is that employees' morale goes up.
Statement II: a systematic approach to safety seeks a quick fix to the solution.
(A) Both Statement I and II are individually true, and Statement II is the correct explanation of Statement I.
(B) Both Statement I and II are individually true, but Statement II is not the correct explanation of Statement I.
(C) Statement I is true, but Statement II is false.
(D) Statement I is false, but Statement II is true.

13. Statement I: SMS and QMS both enhance the safety of an organisation.
Statement II: SMS focuses on the safety of the organisation, whereas QMS focuses on the services and products of the organisation.
(A) Both Statement I and II are individually true and Statement II is the correct explanation of Statement I.
(B) Both Statements I and II are individually true, but Statement II is not the correct explanation of Statement I.
(C) Statement I is true, but Statement II is false.
(D) Statement I is false, but Statement II is true.

14. Statement I: a minimum concentration of 15–16 per cent oxygen in the air is required for the continuation of flaming combustion.
Statement II: the highest concentration of oxygen in the mixture of fuel vapour and oxygen at which ignition takes place is known as Upper Explosive Limit.
(A) Both statements I and II are individually true, and Statement II is the correct explanation of Statement I.
(B) Both statements I and II are individually true, but Statement II is not the correct explanation of Statement I.
(C) Statement I is true, but Statement II is false.
(D) Statement I is false, but Statement II is true.

15. Probability of the event that might occur X Severity of the event if it occurs =
(A) Accident
(B) Hazard
(C) Risk
(D) None of the above

16. Industrial safety management is that branch of management which is concerned with_____hazards from industries.
(A) reducing
(B) controlling
(C) eliminating
(D) All of the above

17. Which of the following is an indirect cost of an accident?
(A) Money paid for treatment of a worker
(B) Compensation paid to a worker
(C) Cost of lost time of an injured worker
(D) All of the above

18. For household wiring and small units, which of the the following should be used for safety measure?
(A) MCB
(B) ACB
(C) OCB
(D) MCCB

19. Decibel (dB) is a unit used to measure:
(A) light
(B) sound
(C) frequency
(D) None of the above

20. Class A fire includes fire due to:
(A) wood
(B) oil
(C) transformer
(D) chemical

Answer Key

1. (D)	**2.** (C)	**3.** (D)	**4.** (B)	**5.** (C)	**6.** (A)	**7.** (A)	**8.** (C)	**9.** (C)	**10.** (D)
11. (C)	**12.** (C)	**13.** (B)	**14.** (C)	**15.** (C)	**16.** (D)	**17.** (C)	**18.** (A)	**19.** (B)	**20.** (A)